

Research Article

Design Vehicle Autonomous System for Lane Change Maneuver

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Design Vehicle Autonomous System for Lane Change Maneuver

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Abstract:

Lane-change maneuvers constitute one of the most safety-critical decision-making tasks in autonomous driving, particularly under highway conditions characterized by high speed and dense traffic. Conventional lane-change strategies often rely on fixed gap thresholds or single surrogate safety measures such as time-to-collision (TTC), which fail to capture the coupled effects of longitudinal dynamics and inter-vehicle spacing. To overcome these limitations, this paper proposes a risk-aware lane-change decision framework based on an adaptive neuro-fuzzy inference system (ANFIS) for SAE Level-3 autonomous vehicles. The proposed method fuses normalized longitudinal acceleration and minimum inter-vehicle gap into a continuous and interpretable risk index ranging from 0 to 1, which is explicitly embedded into a phase-based lane-change decision architecture. Quantitative simulation results obtained in a high-fidelity Rescan–Simulink environment demonstrate that the proposed framework consistently detects hazardous interactions before TTC reaches collision-critical thresholds. In abort-oriented scenarios, the risk index rises beyond the warning activation level while TTC remains recoverable, enabling early maneuver cancellation and subsequent TTC recovery without entering the danger activation region. In contrast, during successful lane-change executions, the risk value remains bounded within low-to-medium ranges (typically below 0.6) throughout the maneuver, indicating sufficient safety margins. Vehicle dynamic responses further confirm smooth maneuver execution, with no abrupt peaks in resultant acceleration or yaw-rate profiles, preserving ride comfort and stability. Compared with threshold-based and learning-driven decision methods, the proposed ANFIS-based framework achieves enhanced risk sensitivity, improved decision stability, lower computational complexity, and full interpretability, making it well suited for real-time autonomous driving applications.

Key words:

Autonomous Vehicles; Lane-Change Decision-Making; Risk Assessment; Adaptive Neuro-Fuzzy Inference System (ANFIS); Time-to-Collision; Risk-Aware Planning; SAE Level-3 Automation

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1. Introduction

Autonomous vehicles (AVs) have emerged as a core component of intelligent transportation systems, offering promising solutions for reducing traffic accidents, improving road safety, and enhancing traffic efficiency. With rapid advancements in sensing, perception, and control technologies, modern AVs are increasingly capable of operating in complex traffic environments with minimal human intervention [1]–[3]. Among various driving maneuvers, lane-changing plays a critical role in highway driving, influencing traffic flow, safety margins, and passenger comfort [4], [5]. Lane-change maneuvers are inherently more complex and risk-prone than longitudinal driving tasks, as they require simultaneous consideration of surrounding vehicles, road geometry, and dynamic constraints [6], [7]. Improper or delayed lane-change decisions are recognized as major contributors to freeway collisions, particularly in dense and high-speed traffic conditions [8]. Consequently, reliable lane-change decision-making and trajectory planning are considered fundamental challenges in autonomous driving systems [9], [10].

Human-driven Lane changes rely heavily on subjective judgment, often resulting in inconsistent safety margins and delayed responses [11]. Traditional rule-based or threshold-based algorithms have been widely adopted in early driver assistance systems; however, their rigid structure limits adaptability under uncertain and rapidly changing traffic scenarios [12], [13]. These methods commonly fail to capture the gradual transition between safe and unsafe situations that characterizes real-world driving behavior [14].

Extensive research has been conducted on lane-change trajectory generation using polynomial curves, Bézier curves, and splines to ensure smooth lateral transitions [15]–[18]. Optimization-based methods incorporating constraints on lateral acceleration, jerk, and vehicle stability have also been proposed to improve ride comfort and safety [19], [20]. Nevertheless, many of these approaches assume that the decision to change lanes has already been made, leaving the decision-making and risk evaluation aspect largely unaddressed [21].

Recent studies have focused on integrating decision-making layers into lane-change frameworks using probabilistic models, utility-based approaches, and game-theoretic formulations [22]–[24]. While these methods offer improved theoretical rigor, they often rely on precise mathematical modeling or extensive parameter tuning, which may not be feasible in highly uncertain real-world traffic environments [25], [26].

Machine learning techniques, including support vector machines, neural networks, and deep reinforcement learning, have been increasingly applied to lane-change decision making [27]–[30]. Although these approaches demonstrate strong performance in data-rich environments, their dependency on large datasets, high computational cost, and limited interpretability pose significant challenges for safety-critical applications such as autonomous driving [31], [32].

Fuzzy logic provides a powerful alternative for modeling uncertainty and human-like reasoning in autonomous driving systems [33], [34]. By representing variables such as gap acceptance and vehicle acceleration through linguistic terms, fuzzy inference systems enable smooth and interpretable decision-making under ambiguous conditions [35]. Several researchers have successfully applied fuzzy logic to collision avoidance and adaptive cruise control, highlighting its suitability for AV environments [36], [37].

Despite notable advancements, a comprehensive framework that jointly evaluates collision risk and lane-change safety using normalized, continuous, and interpretable metrics remains lacking. Many existing methods focus either on trajectory generation or decision-making in isolation, neglecting their strong interdependence [38]–[40]. Furthermore, limited attention has been paid to integrating longitudinal dynamics such as braking intensity with relative gap information in a unified risk assessment model [41].

To address these gaps, this study proposes an intelligent risk-aware lane-change framework based on fuzzy logic inference for autonomous vehicles. Unlike conventional threshold-based or purely data-driven approaches, the proposed method quantitatively evaluates risk by fusing longitudinal acceleration and minimum inter-vehicle distance into a normalized risk index [42]–[44]. This formulation enables smooth transitions between different safety levels, closely resembling human driving perception.

In contrast to optimization-only trajectory planning methods [45], [46], the presented approach incorporates an explicit risk evaluation layer that directly influences lane-change feasibility. Compared with deep-learning-based decision systems [47], [48], the proposed fuzzy framework offers improved interpretability, lower computational complexity, and enhanced suitability for real-time implementation. These characteristics make the approach particularly attractive for safety-critical AV applications [49].

The normalized risk output generated by the proposed framework can be seamlessly integrated into higher-level motion planning and control modules, including emergency braking and adaptive lane-change strategies. This modularity supports scalable deployment across different autonomous driving architectures and traffic scenarios, ranging from highway cruising to congestion mitigation [50].

2. Risk Assessment Based on TTC and Longitudinal Dynamics

Risk assessment in driving and lane-changing scenarios is commonly conducted using Surrogate Safety Measures (SSMs), which estimate collision likelihood and severity without relying on historical crash data. These measures utilize observable kinematic variables such as inter-vehicle distance, relative speed, and acceleration to characterize potentially hazardous situations. Among the most widely used SSMs are Time-to Collision (TTC) and longitudinal acceleration-based indicators.

Time-to-Collision (TTC)

The Time-to-Collision metric represents the remaining time until a collision occurs under the assumption that the current relative motion between two vehicles remains unchanged. TTC is defined based on the longitudinal distance and relative velocity between a following and a leading vehicle:

$$TTC = \frac{D}{\Delta v} = \frac{D}{v_f - v_p}, v_f > v_p \quad (1)$$

where D denotes the longitudinal gap between the two vehicles, and v_f and v_p represent the velocities of the following and leading vehicles, respectively. When both vehicles travel at identical speeds ($v_f = v_p$), TTC tends toward infinity, indicating that TTC alone is insufficient for reliably quantifying collision risk.

Consequently, TTC should be regarded as a necessary but not sufficient safety indicator.

To address the limitations of TTC, a longitudinal acceleration criterion is incorporated to represent the minimum deceleration required by the following vehicle to avoid a collision. Assuming a reduction of the following vehicle's speed to that of the leading vehicle, the required acceleration is expressed as:

$$a_{xc} = \frac{v_p^2 - v_f^2}{2D} \quad (2)$$

A negative value of a_{xc} indicates the necessity for braking. Similar to TTC, this metric also loses effectiveness when the velocities of both vehicles are equal, reinforcing the need for multi-criteria risk evaluation.

In addition to TTC and acceleration metrics, the minimum safe longitudinal distance is employed as a complementary safety indicator. This distance provides an additional safety margin, enabling the execution of maneuvers such as lane changes or emergency braking with reduced collision risk.

3. Fuzzy Logic-Based Risk Inference

To evaluate collision risk during car-following or lane-changing maneuvers, a fuzzy logic-based inference framework is employed to integrate multiple safety-related indicators into a unified decision-making structure. Since collision risk cannot be accurately represented by a single numerical metric, the fuzzy approach enables effective handling of uncertainty, imprecision, and nonlinear interactions among variables. In the proposed system, longitudinal acceleration and minimum inter-vehicle gap is selected as the primary input variables, while the output represents an overall normalized risk level.

In this table 1 longitudinal acceleration variable characterizes braking and acceleration behaviors and is normalized with respect to gravitational acceleration. It is described by three linguistic terms: High (H), Low (L), and Positive (P). Negative values correspond to braking maneuvers, indicating deceleration intensity, whereas positive values represent acceleration actions. This normalization ensures scale consistency and allows the fuzzy system to interpret aggressive or gentle driving behaviors in a meaningful manner.

The minimum gap variable reflects the relative safety margin between the ego vehicle and the preceding vehicle and is normalized with respect to the ego vehicle's velocity. It is defined using three linguistic levels—Low (L), Medium (M), and High (H)—representing unsafe, moderate, and comfortable following distances, respectively. This variable plays a key role in assessing collision risk, particularly in dense traffic or during lane-change operations.

The fuzzy inference system outputs a normalized risk level expressed through five linguistic categories: Very Low (VL), Low (L), Medium (M), High (H), and Very High (VH). These risk levels provide a gradual and interpretable assessment of driving safety rather than a binary safe-unsafe decision. For instance, scenarios involving small headway distances combined with strong braking result in a Very High-risk level, whereas large gaps accompanied by positive acceleration correspond to a Very Low risk state. Medium gap situations combined with mild braking typically yield Medium or Low risk levels, capturing the inherent trade-offs in real driving conditions.

The fuzzy reasoning process follows standard inference steps. First, crisp input values of longitudinal acceleration and minimum gap are mapped onto fuzzy sets using predefined membership functions. Next, fuzzy IF-THEN rules are activated to infer the corresponding risk levels. The outputs of all active rules are then aggregated, and finally, a defuzzification process based on the Center of Gravity (CoG) method is applied to generate a single crisp risk index. This structured inference process ensures smooth transitions between risk states and enhances the interpretability of the decision-making mechanism.

Table 1. Linguistic Variables Used in the Fuzzy Risk Assessment

Variable	Linguistic Term	Normalized Range
Longitudinal Acceleration	High (H)	$[-1, -0.3]$
	Low (L)	$[-0.4, 0.3]$
	Positive (P)	$[-0.1, 1]$
Minimum Gap	Low (L)	$[0.1, 0.6]$
	Medium (M)	$[0.8, 1.8]$
	High (H)	$[1.5, 3]$
Risk Level (Output)	Very Low (VL)	$[0, 0.25]$
	Low (L)	$[0.15, 0.45]$
	Medium (M)	$[0.35, 0.65]$
	High (H)	$[0.55, 0.85]$
	Very High (VH)	$[0.75, 1]$

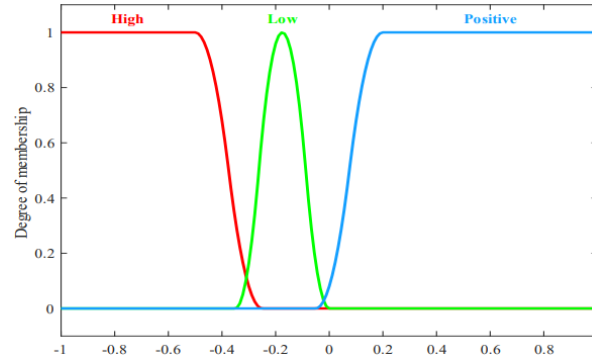


Figure 1. Risk-Adaptive Neuro-Fuzzy Modeling of Braking Behavior for Autonomous Driving Decision-Making

The fuzzy membership functions illustrated in this figure 1 provide a structured and interpretable representation of braking acceleration within the proposed risk-aware autonomous driving framework. By normalizing braking acceleration and categorizing it into three linguistic regions—*High*, *Low*, and *Positive*—the controller is able to map continuous vehicle dynamics into a decision space that reflects both safety and comfort considerations. The *High* braking membership captures emergency deceleration scenarios associated with elevated collision risk, while the *Low* region corresponds to smooth speed adjustment under moderate gap variations. Conversely, the *Positive* membership represents acceleration or risk-free cruising conditions, ensuring that unnecessary braking is avoided. The deliberate overlap between these membership functions enables smooth transitions and prevents abrupt control actions, which is critical in real-world autonomous driving where traffic dynamics evolve continuously. Integrated within the ANFIS structure described in the study, these membership functions allow the control system to adaptively tune braking responses based on real-time risk indicators such as time-to-collision (TTC) and relative velocity. This neuro-fuzzy representation not only enhances robustness against uncertainty and sensor noise but also preserves transparency, addressing a key limitation of black-box learning-based controllers. Consequently, the proposed fuzzy braking model plays a pivotal role in achieving a balanced trade-off between safety, ride comfort, and decision stability in autonomous lane-change and collision-avoidance scenarios discussed in the paper.

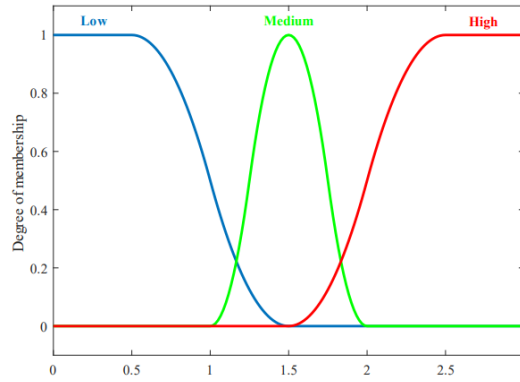


Figure 2. Neuro-Fuzzy Representation of Inter-Vehicle Spacing for Risk-Aware Lane-Change Decisions

The membership functions depicted in this figure 2 model the minimum inter-vehicle gap as a key perceptual variable within the proposed ANFIS-based risk assessment framework for autonomous driving. By normalizing the minimum gap with respect to the following vehicle's velocity, the model captures both spatial and dynamic safety constraints in a unified dimensionless form. The three linguistic categories—*Low*, *Medium*, and *High*—correspond to critically unsafe, transitional, and safe spacing conditions, respectively. The *Low* gap region characterizes imminent collision risk scenarios that require conservative actions such as braking or maneuver suppression, while the *High* gap membership reflects sufficient spatial margins that permit safe lane-change execution. The *Medium* region acts as a decision buffer, enabling gradual adaptation between aggressive and conservative behaviors. The smooth overlap among these membership functions is deliberately designed to avoid abrupt switching in the control logic, thereby ensuring decision continuity and passenger comfort. Within the ANFIS structure, this gap-based fuzzy input interacts with other risk indicators such as relative speed and braking demand, allowing the controller to learn optimal lane-change feasibility boundaries online. As a result, this neuro-fuzzy gap modeling strategy enhances robustness against sensor uncertainty and traffic variability while maintaining interpretability, forming a critical component of the overall intelligent and risk-aware autonomous driving decision system validated in the automotive scenarios of the study.

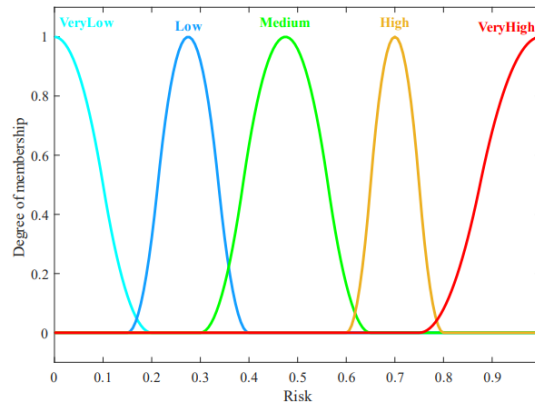


Figure 3. Multi-Level Neuro-Fuzzy Risk Quantification for Autonomous Driving Decision Systems

This figure 3 illustrates a finely granular neuro-fuzzy representation of risk perception designed to support intelligent decision-making in autonomous driving scenarios. The risk variable is modeled using five linguistic membership functions—*Very Low*, *Low*, *Medium*, *High*, and *Very High*—providing a hierarchical and interpretable abstraction of environmental uncertainty and threat intensity. Such multi-level partitioning enables the controller to distinguish subtle transitions between safe, cautionary, and critical situations, which is not possible with coarse or binary risk models. The *Very Low* and *Low* regions correspond to benign traffic conditions where normal cruising or lane-change approval is permissible, while the *medium* region reflects borderline situations requiring adaptive caution and continuous monitoring. The *High* and *Very High* memberships capture increasingly dangerous conditions, such as reduced time-to-collision or insufficient safety gaps, where aggressive maneuvers must be inhibited in favor of braking or evasive control actions. The smooth overlap among adjacent membership functions ensures continuity in risk evaluation, preventing abrupt decision switching and enhancing passenger comfort and system stability. When embedded within the ANFIS framework, this risk modeling layer allows the controller to learn nonlinear risk boundaries online by correlating perceptual inputs (e.g., gap distance, relative velocity, braking demand) with optimal driving responses. Consequently, this neuro-fuzzy risk formulation provides a transparent, adaptive, and robust mechanism for translating complex traffic dynamics into stable autonomous decisions, playing a central role in the successful lane-change and collision-avoidance behaviors validated in the automotive scenarios of the study.

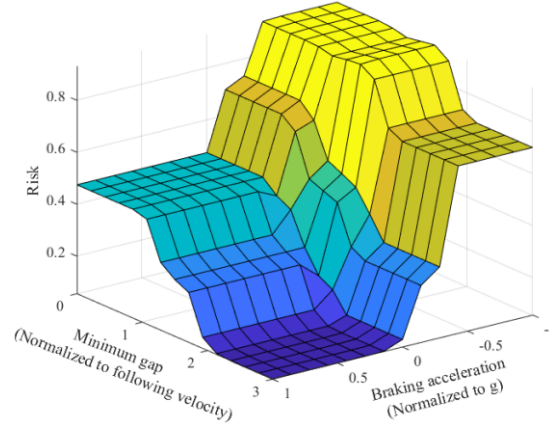


figure 4. Neuro-Fuzzy Risk Surface for Integrated Braking-Gap Decision Modeling in Autonomous Driving

The three-dimensional surface illustrated in this figure 4 represents the learned neuro-fuzzy risk landscape generated by the ANFIS-based decision module as a function of two critical inputs: normalized braking acceleration and normalized minimum inter-vehicle gap. This surface captures the nonlinear coupling between longitudinal control demand and spatial safety constraints, providing a continuous and interpretable mapping from vehicle dynamics to perceived risk. Regions of high risk emerge when strong braking demands coincide with small gap values, reflecting imminent collision scenarios that necessitate conservative or evasive actions. In contrast, low-risk regions correspond to sufficient safety margins combined with mild or positive acceleration, indicating safe conditions for maintaining speed or executing a lane change. The smooth yet nonuniform gradients across the surface highlight the adaptive learning capability of the ANFIS structure, which refines decision boundaries beyond fixed rule-based systems and avoids abrupt switching behavior. Importantly, this risk surface is not heuristically predefined but learned online from error dynamics and rule interactions, allowing the controller to adapt to varying traffic conditions and sensor uncertainty. As a result, the proposed neuro-fuzzy risk surface serves as a core decision layer that enables stable, comfort-oriented, and safety-aware autonomous driving behavior, directly supporting the successful lane-change and collision-avoidance outcomes demonstrated in the automotive validation scenarios of the study.

4. Decision-Making Unit Design for Lane Change Maneuver

Driving in highway environments generally involves lower complexity compared to urban settings, as vehicle motion is more structured and lanes are typically unidirectional. In such scenarios, situational awareness is enhanced through the perception and estimation of surrounding traffic, enabling the ego vehicle to identify neighboring vehicles and assess their accessibility and relative positions.

Based on this information, the risk associated with each surrounding vehicle relative to the ego vehicle is calculated using the fuzzy logic framework introduced in the previous section. These computed risk values are then provided as inputs to the decision-making system.

In this figure 5, a comprehensive decision-making model for highway lane-changing maneuvers is presented for an SAE Level 3 automated vehicle. The proposed approach is capable of generating real-time lane-change requests within a specified time horizon. Upon request, the system evaluates the feasibility of executing or rejecting the lane change and determines the appropriate lane-change direction, denoted as left (LLL) or right (RRR).

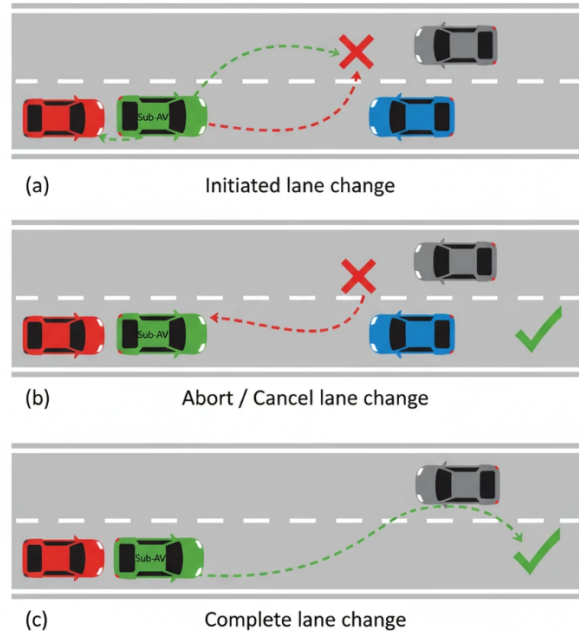


Figure 5. Decision-Making Outcomes of Lane Change Maneuvers in a Subject Autonomous Vehicle under Dynamic Traffic Conditions

The figure 5 illustrates the decision-making outcomes of a lane change maneuver executed by a subject autonomous vehicle (Sub-AV) under varying surrounding traffic conditions, emphasizing the interaction between safety assessment, risk prediction, and trajectory planning. In panel (a), the Sub-AV initiates a lane change based on an initial feasibility evaluation; however, the projected interaction with neighboring vehicles in

the target lane introduces a potential longitudinal and lateral conflict, reflected by the red dashed trajectory and the abort marker. This highlights the limitation of static gap acceptance and underlines the necessity of continuous risk reassessment during maneuver execution. Panel (b) demonstrates the abort or cancel decision, where the Sub-AV actively retreats to its original lane after detecting an unsafe condition, such as insufficient time-to-collision (TTC) or adverse relative velocity profiles with adjacent vehicles. This behavior reflects a conservative safety-oriented policy, prioritizing collision avoidance over maneuver completion. In contrast, panel © represents a successful lane change scenario, where the Sub-AV identifies a dynamically feasible gap and executes a smooth trajectory that satisfies both kinematic constraints and safety margins. Collectively, these scenarios underline the importance of real-time situational awareness and adaptive decision logic in autonomous driving systems, showing that robust lane change performance is not defined solely by successful completion, but by the system's ability to abort or cancel maneuvers when evolving traffic conditions violate predefined risk thresholds.

I. Extended Fuzzy-Based Decision-Making Strategy for Lane Change Maneuvers

Lane-change warning generation in the proposed framework is not restricted solely to the second lane-change region. Depending on traffic conditions, warning issuance may also be triggered within Region I, provided that the predefined risk thresholds are satisfied.

In the previous section, a fuzzy logic system was employed to transform environmental state variables—namely longitudinal deceleration (braking) and safe longitudinal distance—into corresponding risk values. This risk estimation was performed individually for each surrounding object in the vicinity of the autonomous vehicle (AV), enabling continuous assessment of collision likelihood.

In the present section, a new fuzzy logic module is introduced, whose inputs are the risk levels of surrounding vehicles, while its outputs directly govern lane-change decision-making and longitudinal control signals. Importantly, this inference structure always incorporates **three vehicles** in the fuzzy reasoning process:

- the preceding vehicle in the current lane,
- the preceding vehicle in the target lane,
- the following vehicle in the target lane.

It is assumed that the following vehicle in the current lane maintains a safe distance and appropriate speed throughout the maneuver and therefore does not directly influence the decision process.

I. Decision Outputs and Control Actions

The first output of the fuzzy inference system determines the lane-change decision status, i.e., whether the maneuver should be executed, delayed, or aborted. This output is selected in accordance with predefined decision rules illustrated in Figure 6 and summarized in Table 2.

The second output corresponds to the longitudinal braking command, which is dedicated to speed control during the maneuver. The fuzzy rules governing this output are presented.

The complete set of fuzzy rules that map the risk levels of the three surrounding vehicles to the lane-change decision state is provided in Table 2.

The decision-making strategy applied in this study is explicitly programmed based on spatial segmentation of the lane-change process. As shown in Figure 6, the lane-change area is divided into two consecutive regions, separated by a midpoint that corresponds precisely to the intersection between the planned lane-change trajectory and the reference centerline of the target lane.

After initiating a lane-change maneuver in Region I, the system is capable of either:

- completing the maneuver as planned, or
- canceling the maneuver and returning to the original lane at any intermediate point along the path, based on updated risk evaluations.

Once the vehicle passes the midpoint (point b) and enters Region II, it is committed to the lane-change trajectory. In this region—particularly near the end of the lane-change curve—returning to the original lane introduces a high risk of collision and significantly complicates vehicle control.

To mitigate this risk, the system aims to finalize the maneuver robustly, while simultaneously monitoring environmental conditions. If necessary, the decision unit may request driver intervention, ensuring safety compliance within SAE Level 3 autonomous driving constraints.

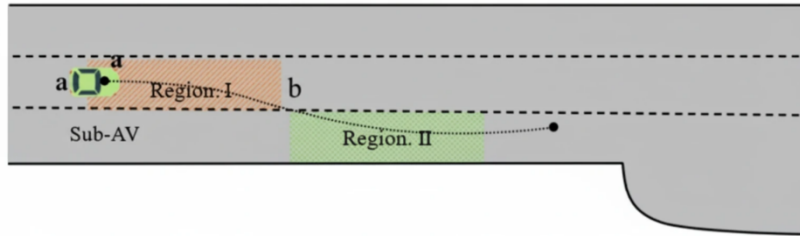


Figure 6. Decomposition of Lane Change Trajectory into Sequential Decision Regions for the Subject Autonomous Vehicle

The figure 6 presents a structured decomposition of the lane change maneuver of the subject autonomous vehicle (Sub-AV) into two sequential decision regions, enabling a clearer interpretation of maneuver planning and execution. Region I represents the initial decision and preparation phase, where the Sub-AV evaluates the feasibility of lane change based on local traffic conditions, relative vehicle states, and safety constraints, while maintaining its current lane. During this phase, lateral motion is minimal and the vehicle primarily focuses on longitudinal gap assessment. The transition point (b) marks the commitment boundary, at which a feasible and dynamically safe gap is identified, and the maneuver advances to Region II. Region II corresponds to the execution phase, where coordinated lateral and longitudinal control actions are applied to realize a smooth trajectory toward the target lane while satisfying kinematic continuity and safety margins. This regional segmentation highlights the hierarchical nature of autonomous lane change decision-making, demonstrating that successful maneuver execution relies not only on trajectory optimization, but also on explicit phase-based reasoning that allows early termination, refinement, or continuation of the maneuver under evolving traffic conditions.

This table 2 a rule-based decision framework for determining the feasibility of a lane change maneuver based on the quantified risk levels of surrounding vehicles. The decision process incorporates three independent risk sources: the adjacent lane vehicle (Risk A), the rear vehicle in the current or target lane (Risk B), and the destination lane vehicle (Risk D), each categorized into five linguistic risk levels ranging from Very Low (VL) to Very High (VH). The output of the matrix is a binary decision variable, where a value of 1 denotes permission to initiate or continue the maneuver, and -1 represents maneuver rejection or cancellation. The structure of the table reflects a conservative safety-oriented policy, whereby increasing risk in any critical dimension—particularly destination lane risk—progressively restricts maneuver execution. Notably, the decision boundaries emphasize that even moderate increases in surrounding vehicle risk can override favorable conditions elsewhere, thereby preventing unsafe commitment. This rule-based formulation enhances interpretability and robustness, offering a transparent mechanism to integrate multi-vehicle risk perception into autonomous lane change decision-making, while maintaining consistency with human-inspired cautious driving behavior.

Table 2. Rule-Based Decision Matrix for Lane Change Maneuver Considering Surrounding Vehicle Risk Levels

		Risk (A)	Risk (D)				
			<i>VL</i>	<i>L</i>	<i>M</i>	<i>H</i>	<i>VH</i>
Risk (B)	<i>VL</i>	<i>VL</i>	1	1	1	-1	-1
		<i>L</i>	1	1	1	-1	-1
		<i>M</i>	1	1	1	-1	-1
		<i>H</i>	-1	-1	-1	-1	-1
		<i>VH</i>	-1	-1	-1	-1	-1
	<i>L</i>	<i>VL</i>	1	1	1	-1	-1
		<i>L</i>	1	1	1	-1	-1
		<i>M</i>	1	1	1	-1	-1
		<i>H</i>	-1	-1	-1	-1	-1
		<i>VH</i>	-1	-1	-1	-1	-1
	<i>M</i>	<i>VL</i>	1	1	1	-1	-1
		<i>L</i>	1	1	1	-1	-1
		<i>M</i>	1	1	1	-1	-1
		<i>H</i>	-1	-1	-1	-1	-1
		<i>VH</i>	-1	-1	-1	-1	-1
	<i>H</i>	<i>VL</i>	1	1	1	-1	-1
		<i>L</i>	1	1	1	-1	-1
		<i>M</i>	1	1	1	-1	-1
		<i>H</i>	-1	-1	-1	-1	-1
		<i>VH</i>	-1	-1	-1	-1	-1
	<i>VH</i>	<i>VL</i>	-1	-1	-1	-1	-1
		<i>L</i>	-1	-1	-1	-1	-1
		<i>M</i>	-1	-1	-1	-1	-1
		<i>H</i>	-1	-1	-1	-1	-1
		<i>VH</i>	-1	-1	-1	-1	-1

The figure 7 depicts an integrated system-level architecture for lane change decision-making and control transition in an automated driving system operating under defined Operational Design Domain (ODD) conditions. The framework begins with route planning and continuous evaluation of ODD validity and ADS activation status, ensuring that autonomous functions are only executed within permissible operational bounds. Environmental perception and connectivity modules, including on-board sensors (cameras, radars, GPS/maps) and V2 communications, provide situational awareness to the high-level decision-making layer. This layer performs strategic and tactical reasoning, such as highway pilot and traffic jam driving, and generates lane change requests in accordance with standardized automation definitions (e.g., NHTSA). Upon receiving a maneuver request, the Lane Change Management Unit evaluates feasibility and safety, followed by path-planning and re-planning to generate dynamically consistent trajectories. Control commands are then issued to the steering, braking, and throttle systems, ultimately influencing vehicle dynamics. In parallel, a take-over management unit continuously monitors time-critical risk indicators, such as time-to-collision and remaining maneuver time, particularly during system failures or ODD exits. A warning system mediates interaction with the human driver whenever automation limits are reached, enabling a structured and safe transition of control. Overall, the architecture demonstrates a hierarchical and human-aware decision loop, where autonomous maneuver execution, safety supervision, and driver engagement are tightly coupled to ensure robustness under complex and evolving traffic scenarios.

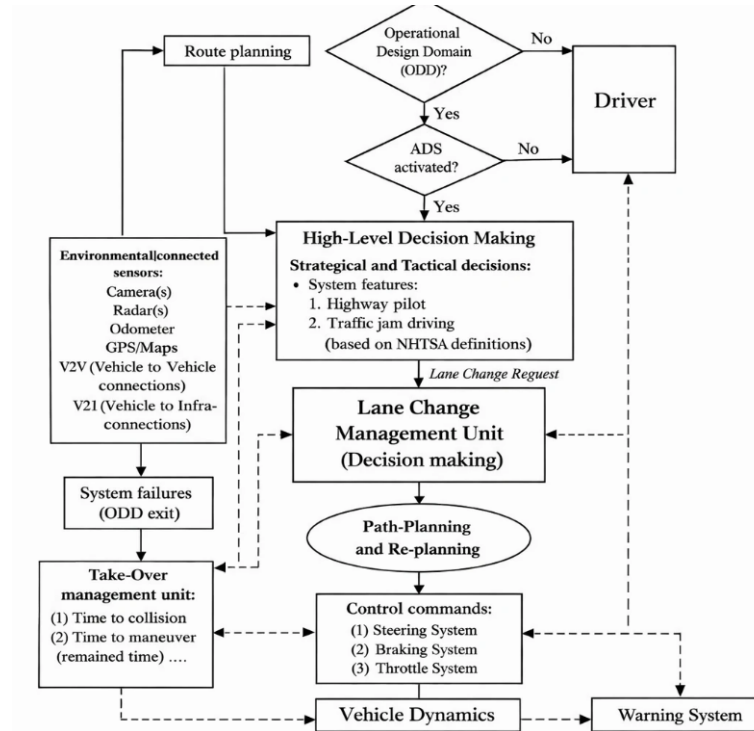


Figure 7. Integrated Decision-Making Architecture for Lane Change and Take-Over Management in Automated Driving Systems

5. Decision-Making Unit Design for Lane Change Maneuver

Driving in highway environments generally involves lower complexity compared to urban settings, as vehicle motion is more structured and lanes are typically unidirectional. In such scenarios, situational awareness is enhanced through the perception and estimation of surrounding traffic, enabling the ego vehicle to identify neighboring vehicles and assess their accessibility and relative positions. Based on this information, the risk associated with each surrounding vehicle relative to the ego vehicle is calculated using the fuzzy logic framework introduced in the previous section. These computed risk values are then provided as inputs to the decision-making system. In this figure 8, a comprehensive decision-making model for highway lane-changing maneuvers is presented for an SAE Level 3 automated vehicle. The proposed approach is capable of generating real-time lane-change requests within a specified time horizon. Upon request, the system evaluates the feasibility of executing or rejecting the lane change and determines the appropriate lane-change direction, denoted as left (L) or right (R).

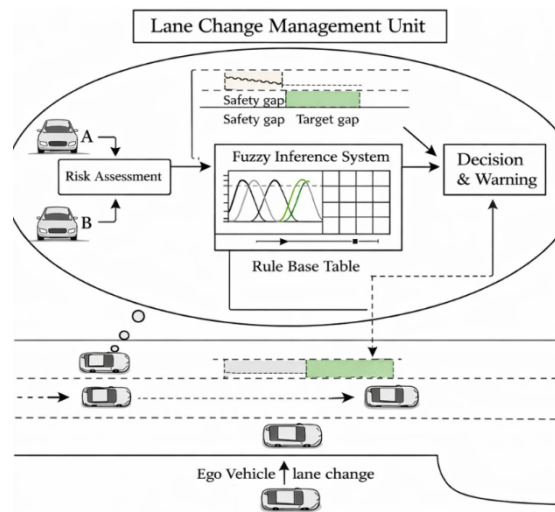


Figure 8. Schematic architecture of the Lane Change Management Unit for SAE Level-3 automated driving

In the figure 8 risk levels of surrounding vehicles (A and B) are first evaluated and mapped into linguistic variables using a fuzzy inference system. Safety and target gaps are analyzed to assess lane availability, while fuzzy membership functions and a rule base table generate the final lane-change decision and warning. The lower part illustrates a highway scenario in which the ego vehicle performs a lane-change maneuver toward a safe target gap.

SAE Level 3 automated vehicles are inherently classified as conditionally automated driving systems, meaning that their autonomous functionality is restricted to a predefined set of operational design domain (ODD) conditions. These conditions may be influenced by factors such as road type,

traffic density, environmental and weather conditions, as well as system performance limitations. Consequently, the continuous presence of a human driver and their readiness to intervene at any moment are essential to ensure a safe transition of control whenever the automated system reaches its operational limits.

Under such circumstances, the Take-Over Request (TOR) mechanism is activated. This mechanism is responsible for informing the driver of the automated system's inability to safely continue driving and for managing the transition of control from automated to manual mode. Since drivers may engage in secondary non-driving-related tasks during automated driving, their readiness level prior to receiving a TOR is typically reduced, necessitating a finite period for attention recovery and control resumption.

Previous studies have demonstrated that drivers require a non-negligible time interval to re-adapt to the driving task and achieve a stable driving state, which is defined as the condition in which the driver can operate the vehicle in accordance with their normal driving style and control behavior. Therefore, the design of the take-over management unit must explicitly account for an adequate transition duration to ensure safety.

Reports published by the National Highway Traffic Safety Administration (NHTSA) indicate that driver reaction times vary significantly across different scenarios. For SAE Level 3 vehicles, the average reaction time to an urgent warning is approximately 2.3 seconds, whereas staged warning strategies may result in longer response times. Additional studies have estimated the overall take-over time under normal conditions to be approximately **4 seconds**.

A take-over request may be triggered by multiple factors, including the vehicle approaching the boundaries of its operational design domain, the emergence of unexpected traffic situations, adverse weather conditions, sensor malfunctions, the inability of the decision-making system to generate a safe response, or sudden failures such as radar loss or tire blowouts.

To temporally analyze the take-over process, three key time intervals are defined:

- T_1 : the driver's initial reaction time to the take-over warning
- T_2 : the time required for the driver to re-assume control of the vehicle
- T_3 : the time needed for the driver to reach a stable driving state

If these time intervals are inadequately calibrated, the likelihood of unsafe driving behaviors and potential collisions increases.

In this study, take-over warnings during lane-change maneuvers are classified into two categories: urgent warnings and staged warnings. In urgent warning scenarios, the driver is required to regain control within the shortest possible time, whereas staged warnings provide an extended transition window, gradually preparing the driver for intervention. The selection of an appropriate warning strategy is determined by the instantaneous risk level and prevailing driving conditions.

The figure 9 illustrates the functional architecture of a Take-Over Management Unit (TOMU) designed to manage the transition of control from automated driving to the human driver in conditionally automated vehicles. The framework integrates multi-layer information sources, including driver state monitoring (e.g., eye gaze, hand position, and distraction), vehicle risk metrics such as time-to-collision (TTC), required deceleration, and minimum gap, as well as environmental and Operational Design Domain (ODD) status. These inputs are processed by a decision logic module that jointly evaluates driver readiness and situational risk, forming the basis for takeover necessity assessment. Subsequently, a time budget assessment module estimates the available response time, determining whether sufficient temporal margin exists for safe human intervention. The staged evaluation enables adaptive takeover requests, ensuring that control transition is not solely event-driven but context-aware and human-centric. The lower timeline depiction emphasizes the gradual and managed shift from automated driving to manual control, highlighting that successful takeover performance depends on synchronized assessment of driver capability, environmental risk, and temporal constraints. This architecture improves safety and robustness by minimizing abrupt control handovers and reducing the likelihood of inadequate driver response during critical scenarios.

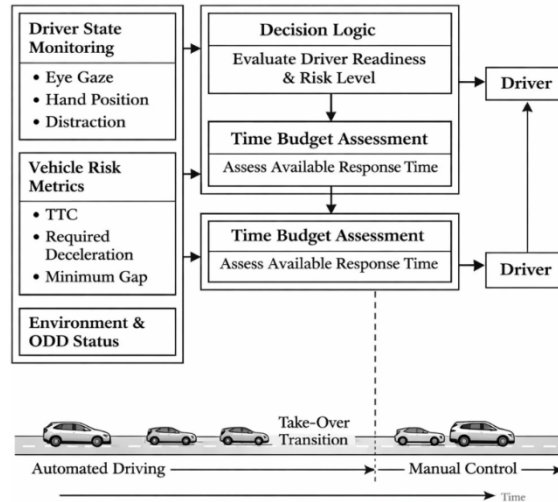


Figure 9. Architecture of the Take-Over Management Unit for Human-Automation Transition in Automated Driving

In this figure 9, lane geometry is represented using virtual paths rather than being directly measured from the road structure. Various techniques exist for generating continuous trajectories, among which Bézier curves are widely adopted due to their smoothness and controllability. One of the key advantages of Bézier curves is their strong geometric control properties, which make them particularly suitable for trajectory planning applications.

In this work, the path generation strategy is based on control-point-driven Bézier curve modeling, following the methodology presented in previous studies. A fifth-order Bézier curve is employed, and its general formulation is expressed as:

$$f_{x,y} = \sum_{i=0}^5 \binom{5}{i} (1-t)^{5-i} t^i P_{i,x,y}, 0 \leq t \leq 1 \quad (3)$$

By expanding the above equation, the Bézier curve can be written as:

$$f_{x,y} = (1-t)^5 P_0 + (1-t)^4 t P_1 + (1-t)^3 t^2 P_2 + (1-t)^2 t^3 P_3 + (1-t) t^4 P_4 + t^5 P_5 \quad (4)$$

where $f_{x,y}$ represents the parametric trajectory, and $P_{i,x,y}$ denote the coordinates of the Bézier control points.

The selected Bézier curve consists of six control points. These control points are defined such that the first point coincides with the starting position of the lane-change maneuver, while the last point corresponds to the final target position after lane transition. For simplicity and robustness, the spacing between consecutive control points is assumed to be uniform. Figure X illustrates the Bézier-based lane-change trajectory along with the distribution of control points.

Given the known initial and terminal positions, the remaining control points can be systematically determined, resulting in a fully defined lane-change trajectory. Accordingly, the path planning module focuses on determining the longitudinal displacement of the terminal point, denoted as x_f .

To ensure passenger comfort and vehicle stability, the trajectory is optimized under a maximum lateral acceleration constraint. The maximum curvature of the Bézier curve is computed using:

$$k_{\text{Bezier,max}} = \frac{a_{y,\text{max}}}{v^2} \quad (5)$$

By substituting the Bézier curvature formulation into the above constraint, the following nonlinear equation is obtained:

$$\left(1 + \left(\frac{2.688}{x_f}\right)^2\right)^{1.5} - \frac{20.2}{x_f^2} = 0 \quad (6)$$

Solving this equation yields the optimal value of x_f , which directly determines the trajectory length and curvature characteristics. Consequently, the complete parametric representation of the Bézier lane-change trajectory, along with its maximum curvature profile, can be explicitly derived. The figure z demonstrates the process of lane change trajectory generation in intelligent vehicles using a fifth-order Bézier curve, where a set of six optimally distributed control points defines a smooth and collision-free path. By parameterizing the curve with key points corresponding to the start position, intermediate transitions, and the final target within the lane boundaries, the method ensures continuous position, heading, and curvature profiles throughout the maneuver. This geometric design inherently satisfies both safety and comfort constraints by minimizing maximum lateral acceleration and ensuring the vehicle remains fully within the drivable corridor. The systematic placement of intermediate control points enables precise modulation of curvature, thus directly influencing ride quality and vehicle stability during the maneuver. Furthermore, the use of a Bézier-based path facilitates low computational cost and deterministic behavior, making it well-suited for real-time deployment in SAE Level-3 autonomous driving systems. Such a trajectory serves as a foundation for integration with higher-level decision units—such as fuzzy logic-based Lane Change Management modules—where kinematic and risk assessment criteria jointly dictate the feasibility and timing of automated lane transitions. The approach, therefore, provides an effective and analytically tractable solution for ensuring robust, safe, and comfortable lane changes in advanced driver-assistance and autonomous driving platforms.

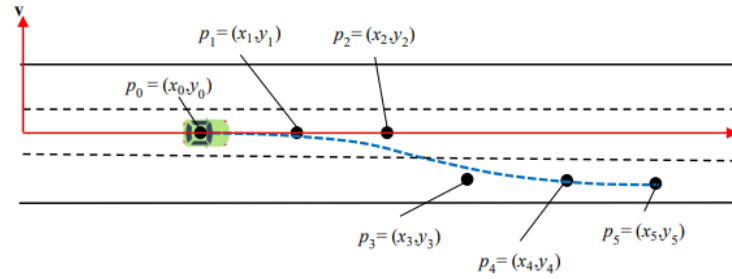


Figure 10. Bézier-Based Lane Change Trajectory with Control Point Distribution

In this figure 10, a six-degree-of-freedom (6-DOF) vehicle dynamic model is employed to describe the coupled translational and rotational motions of the vehicle body. The translational motions occur along the longitudinal x , lateral y , and vertical z axes, while the rotational motions consist of roll (ϕ), pitch (θ), and yaw (ψ) dynamics.

The vehicle-fixed coordinate system, applied forces, and moments are illustrated in Figure 10. Although the vehicle is inherently modeled as a two-track system, additional rotational motions (roll and pitch) are incorporated to extend the model beyond the traditional bicycle approximation.

- The longitudinal motion of the vehicle is governed by the force equilibrium along the x -axis.
- The longitudinal acceleration is expressed as the net longitudinal force divided by vehicle mass:

$$\frac{\sum F_x}{m} = \dot{x} \quad (7) \text{The total longitudinal force is composed of the driving}$$

torque and resistive forces:

$$\sum M_x = mh_{CG}[\dot{\psi}\cos(\phi) + g\sin(\phi) + v_x\dot{\psi}\cos(\phi)] - K_{\text{roll}}\phi - D_{\text{roll}}\dot{\phi} \quad (8)$$

Here, h_{CG} denotes the height of the center of gravity, while K_{roll} and D_{roll} represent roll stiffness and damping.

Pitch motion is described by the moment balance around the lateral axis.

$$\frac{\sum M_y}{I_{yy}} = \dot{\theta} \quad (9)$$

$$\sum M_y = l_f[K_f(z - l_f\theta) + D_f(\dot{z} - l_f\dot{\theta})] - l_r[K_r(z + l_r\theta) + D_r(\dot{z} + l_r\dot{\theta})] - mh_{CG}\dot{x} \quad (10)$$

Yaw motion is governed by the lateral tire forces acting at the front and rear axles.

$$\frac{\sum M_z}{I_{xz}} = \dot{\psi} \quad (11)$$

$$\sum M_z = F_{yf}\cos(\delta)l_f - F_{yr}l_r \quad (12)$$

where l_f and l_r are the distances from the center of gravity to the front and rear axles.

After designing the reference path, the lateral control problem is formulated using the state vector:

$$\mathbf{x} = [e_1, \dot{e}_1, e_2, \dot{e}_2]^T \quad (13)$$

where e_1 is the lateral position error and e_2 is the heading angle error.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_1u + \mathbf{B}_2\text{sitcs} \quad (14)$$

The system matrices are defined as:

$$A = \begin{bmatrix} 0 & \frac{1}{2C_r + 2C_r} & \frac{0}{2C_r + 2C_r} & -\frac{0}{2C_r L_r + 2C_r \cdot 4} \\ 0 & -\frac{m+4}{m+4} & \frac{m}{m} & -\frac{m+4}{m+4} \\ 0 & \frac{0}{2C_r L_r - 2C_r \cdot} & \frac{0}{2C_r L_r - 2C_r \cdot 5} & -\frac{1}{2C_r L_r + 2C_r \cdot \xi} \\ 0 & -\frac{L_s}{L_s} & \frac{L_s}{L_s} & -\frac{L_s}{L_s} \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 0 \\ \frac{2G_t}{m} \\ 0 \\ \frac{2G_{th}}{L} \end{bmatrix}$$

$$B_2 = \begin{bmatrix} 0 \\ \frac{2C_0 l - 2C_0 - l}{mm} \\ 0 \\ \frac{2C_0 l + 2C_0 - \ell}{L_0 u} \end{bmatrix}$$

A state-feedback controller is applied:

$$u = -K\mathbf{x} + \delta \quad (15)$$

Substituting into the state equation yields the closed-loop dynamics:

$$\dot{\mathbf{x}} = (A - B_1 K)\mathbf{x} + B_1 \delta + B_2 i_{des} \quad (16)$$

Due to the coupling between lateral and heading errors, ϵ_1 cannot converge to zero solely through feedback; therefore, an additional steering compensation term is introduced.

The steering angle required for path tracking is defined as:

$$\delta = \tan^{-1} \left(\frac{l_f + l_4}{l_f} \tan(\sin^{-1}(l_4 \kappa_v)) \right) \quad (17)$$

Unlike conventional line-based lane change methods, this study adopts a Bézier curve-based trajectory generation approach. The lateral terminal position y_f is defined as a function of the Bézier parameter τ , which evolves from 0 (start of maneuver) to 1 (end of maneuver).

$$y_{f,ncw} = \begin{cases} y_{f,old} \cdot \frac{\tau}{L}, & \tau < 1 \\ y_{f,old}, & \tau \geq 1 \end{cases} \quad (18)$$

This formulation enables a smooth and comfortable lane-change maneuver while maintaining continuous curvature and bounded lateral acceleration.

This table 3 defines the linguistic structure of the brake pressure output used in the fuzzy inference layer of the proposed autonomous vehicle control system. By expressing brake pressure in terms of four interpretable linguistic values—*Negative*, *Low*, *Medium*, and *High*—the controller establishes a transparent bridge between numerical actuator commands and human-understandable braking behavior. The inclusion of a *Negative* brake pressure region enables the system to explicitly represent release or counter-braking actions, which is essential for smooth transitions between acceleration and deceleration phases. The *Low* and *Medium* ranges correspond to comfort-oriented braking typically associated with adaptive speed regulation and precautionary risk mitigation, while the *High* region captures aggressive braking required in critical safety scenarios. The intentional overlap among numerical ranges ensures continuity in control output, preventing abrupt changes in brake pressure that could compromise vehicle stability or passenger comfort. Within the ANFIS-based decision framework, these linguistic brake pressure outputs are adaptively selected based on real-time risk evaluation and perceptual inputs, allowing the controller to modulate braking intensity according to traffic dynamics and uncertainty levels. As a result, this fuzzy-based brake pressure formulation enhances robustness, interpretability, and smooth actuator engagement, forming a key component of the risk-aware longitudinal control strategy validated in the autonomous driving scenarios of the study.

Table 3. Fuzzy Linguistic Representation of Brake Pressure Output for Autonomous Vehicle Control

Linguistic Value	Notation	Numerical Range
Negative	N	[-20 to 20]
Low	L	[10 to 45]
Medium	M	[40 to 80]
High	H	[70 to 110]

This table 4 presents the complete fuzzy inference rule base employed for extracting the brake control signal according to the perceived risk posed by surrounding vehicles. The rule matrix integrates three risk inputs—**Risk (A)**, **Risk (B)**, and **Risk (D)**—each described using five linguistic levels (*Very Low (VL)*, *Low (L)*, *Medium (M)*, *High (H)*, *Very High (VH)*). The output linguistic decision, denoted by *N*, *L*, *M*, and *H*, represents the required braking action intensity, ranging from no braking to aggressive braking.

The structure of the rule base reflects a safety-oriented design philosophy in which braking intensity increases monotonically with the overall risk level. When all surrounding risks are low or negligible (VL–L), the controller predominantly outputs *N*, indicating no braking action to preserve ride comfort and efficiency. As the combined risk transitions to moderate levels (M), the rules consistently activate *L*-level braking, enabling precautionary deceleration without inducing abrupt longitudinal dynamics. For high and very high-risk combinations (H–VH), the rule base enforces *M* to *H* braking outputs across all distance-risk states, emphasizing collision avoidance and safety prioritization over comfort.

Notably, the symmetry and consistency of the rule distribution prevent contradictory decisions and eliminate rule conflicts, ensuring robust and predictable behavior under varying traffic conditions. This formulation allows the ANFIS-based decision layer to smoothly interpolate between rules, producing continuous brake commands while retaining full interpretability—an essential requirement for safety-critical autonomous driving systems. Consequently, this fuzzy rule base serves as the core logical mechanism that links multi-vehicle risk perception to stable, risk-aware longitudinal control, directly supporting the collision avoidance and lane-change validation scenarios demonstrated in the study.

Table 4. Fuzzy Rule Base for Brake Signal Extraction Based on Surrounding Vehicle Risk Levels

		Risk (A)	Risk (D)				
			<i>VL</i>	<i>L</i>	<i>M</i>	<i>H</i>	<i>VH</i>
Risk (B)	<i>VL</i>	<i>VL</i>	N	N	N	N	N
		<i>L</i>	N	N	N	N	N
		<i>M</i>	L	L	N	N	N
		<i>H</i>	N	N	N	N	N
		<i>VH</i>	N	N	N	N	N
	<i>L</i>	<i>VL</i>	N	N	N	N	N
		<i>L</i>	N	N	N	N	N
		<i>M</i>	L	L	N	N	N
		<i>H</i>	N	N	N	N	N
		<i>VH</i>	N	N	N	N	N
	<i>M</i>	<i>VL</i>	L	L	L	L	L
		<i>L</i>	L	L	L	L	L
		<i>M</i>	L	L	L	L	L
		<i>H</i>	L	L	L	L	L
		<i>VH</i>	L	L	L	L	L
	<i>H</i>	<i>VL</i>	M	M	M	M	M
		<i>L</i>	M	M	M	M	M
		<i>M</i>	M	M	M	M	M
		<i>H</i>	M	M	M	M	M
		<i>VH</i>	M	M	M	M	M
	<i>VH</i>	<i>VL</i>	H	H	H	H	H
		<i>L</i>	H	H	H	H	H
		<i>M</i>	H	H	H	H	H
		<i>H</i>	H	H	H	H	H
		<i>VH</i>	H	H	H	H	H

6. Scenario Design and Simulation Environment

The driving scenarios in this study are developed using the Rescan simulation platform, which provides an integrated and comprehensive environment for modeling vehicle dynamics, sensor configurations, and traffic interactions. Rescan enables seamless coupling with Simulink through a dedicated compilation interface, allowing efficient implementation and evaluation of advanced control and decision-making algorithms. Within this framework, the Rescan graphical user interface is primarily employed for scenario construction and configuration, while dynamic entities and simulation execution are managed through automatically generated model blocks.

All dynamic objects within the simulated environment are represented using the Rescan Actors toolbox, which supports a wide range of traffic participants, including passenger vehicles, trucks, motorcycles, bicycles, and pedestrians. Each actor can be independently configured in terms of vehicle dynamics, driver behavior, physical characteristics, and predefined or observed motion trajectories. Once these actors are incorporated into the graphical environment, dedicated simulation blocks are automatically created, enabling direct integration of control algorithms within the Simulink platform without the need for manual coding of vehicle interactions.

To enable realistic perception of the driving environment, the Rescan Sensors toolbox is utilized to provide a variety of onboard sensing modules, including radar, LiDAR, ultrasonic sensors, and camera systems. These sensors can be attached to any actor in the scenario and individually configured with specific parameters and customized signal-processing pipelines. Such sensor modeling plays a crucial role in reproducing realistic driving conditions and ensuring accurate perception of surrounding traffic and environmental risks.

The overall objective of the scenario design is to create a highway driving environment in which the subordinate autonomous vehicle (Sub-AV) is required to perform a lane-change maneuver. As discussed previously, one of the key capabilities expected from an SAE Level-3 autonomous vehicle is the ability to independently execute lane-change maneuvers on highways and expressways. Accordingly, the Sub-AV must autonomously identify an appropriate exit location, determine the required number of lane changes, and perform the maneuver safely and efficiently. These considerations form the core foundation of the scenario design and simulation environment employed in this research.

7. Scenario Assumptions and Traffic Conditions

Three distinct driving scenarios are designed in this study, each defined by different initial conditions, traffic flow densities, and dynamic interactions among surrounding vehicles. Traffic conditions are modeled as time-varying to realistically reflect changes encountered in highway environments. At the beginning of each scenario, the lane-change system receives a command from the autonomous driving module, initiating both trajectory planning and maneuver execution. Throughout the maneuver, the lane-change unit continuously monitors surrounding traffic conditions, evaluates the associated risk levels, and determines whether the maneuver should be completed as planned, modified, or aborted to ensure safety.

To enhance safety, a warning and risk management strategy is incorporated into the proposed framework. When the evaluated risk level increases during the lane-change maneuver without reaching a critical threshold, an early warning is issued to indicate elevated but manageable risk. A critical warning is triggered if the maneuver cannot be completed within a predefined time window, particularly when the vehicle is approaching a highway exit. In the absence of driver intervention following these warnings, the autonomous system responds by either executing a minimum-risk maneuver under critical conditions or safely decelerating and stopping the vehicle when continuation of the lane-change maneuver is deemed infeasible. It is worth noting that manual takeover or driver return is not considered an objective of this work; instead, system safety is always prioritized over maneuver completion.

The road environment and static infrastructure are modeled using the infrastructure toolbox, which enables the construction of realistic highway scenarios. This toolbox is employed to define straight and curved road segments, highway entrances and exits, intersections and roundabouts, as

well as lane markings, boundaries, traffic signs, and surrounding buildings. Such detailed infrastructure modeling ensures that the simulated scenarios closely resemble real-world driving conditions and provides a reliable platform for evaluating the proposed lane-change and risk-management strategies in complex traffic environments.

8. Highway Geometry and Initial Conditions

For all lane-change scenarios considered in this study, a straight three-lane highway is constructed, including one exit and a connecting on-ramp to realistically represent freeway driving conditions. The total length of the highway is set to 1500 m, and a reference coordinate system with its origin located at (0, 0) is defined at the beginning of the road centerline.

At the start of each simulation, the subordinate autonomous vehicle (Sub-AV) is initially positioned in the center lane at a longitudinal distance of 300 m from the beginning of the highway. Under this initial configuration, the remaining distance from the vehicle to the highway exit is 700 m. The primary objective of the maneuver is to execute a right-lane change into the adjacent lane, which is defined with a standard lane width of 3.75 m.

9. Vehicle Models and Traffic Participants

In each driving scenario, the subordinate autonomous vehicle (Sub-AV) operates in the presence of surrounding traffic vehicles, whose longitudinal positions, velocities, and accelerations vary dynamically throughout the simulation to reflect realistic highway conditions. The Audi A8 sedan is selected as the reference autonomous vehicle platform due to its well-documented dynamic behavior and suitability for automated driving research. The dynamic and physical characteristics of this vehicle, including mass properties, yaw inertia, geometric distances between the center of gravity and axle locations, tire cornering stiffness, steering ratio, steering limits, and acceleration constraints, are specified to ensure accurate vehicle modeling. These parameters are adopted consistently across all scenarios to provide a reliable and reproducible foundation for evaluating the proposed lane-change and decision-making strategies.

The dynamic and physical parameters of the vehicle model used in this study are defined to accurately represent real driving behavior in highway scenarios. The vehicle mass is set to 1820 kg, while the yaw moment of inertia is specified as $3746 \text{ kg}\cdot\text{m}^2$, capturing the rotational characteristics of the vehicle body. The distances from the vehicle's center of gravity to the front and rear axles are defined as 1.17 m and 1.77 m, respectively, ensuring a realistic longitudinal mass distribution. Tire dynamics are modeled through the cornering stiffness parameters, with values of 72 653 N/rad for the front tires and 121 449 N/rad for the rear tires, allowing accurate representation of lateral force generation during steering maneuvers. The steering system is characterized by a steering ratio of 20 and a maximum steering wheel angle of 500 degrees, defining the physical limits of steering actuation. Furthermore, longitudinal motion constraints are incorporated by setting the maximum allowable acceleration to 0.3 g, while the maximum deceleration is limited to g. These parameters are consistently applied across all simulations to ensure reliable and reproducible evaluation of the proposed autonomous lane-change strategy.

10. Sensor Modeling and Configuration

After completing the scenario construction phase, the next step involves equipping the autonomous vehicle with perception sensors suitable for SAE Level 3 automated driving. In the Rescan simulation environment, two distinct sensor modeling approaches are available: ideal sensors and realistic sensors.

Ideal sensors are not affected by object detection uncertainties and assume perfect perception performance. In contrast, realistic sensors are subject to measurement noise, occlusions, and misclassification, meaning that an obstacle may be incorrectly detected or entirely missed. In this study, only the Sub-AV vehicle is equipped with realistic sensors, while the surrounding vehicles follow predefined trajectories based on assigned speed and acceleration profiles from the beginning to the end of the simulation.

In this section, the configuration of six different radar sensors and a forward-looking camera for the vehicle was introduced. Each sensor in the Rescan environment possesses a set of configurable parameters, including mounting position, orientation angles, and intrinsic characteristics.

Key sensor parameters include detection range, field of view, operating frequency, maximum detectable objects, resolution, antenna configuration, and noise characteristics. In this section, the mounting location and parameter settings of each sensor are explicitly defined and adjusted.

Figure 11 illustrates the lateral placement and sensing boundaries of the sensors as visualized from the top view in the Rescan environment. As an example, one long-range radar is installed at both the front and rear of the vehicle. The configuration settings shown in Figure 11 are applied individually to each sensor.

This section describes the sensor modeling strategy adopted for SAE Level 3 autonomous driving simulation using Rescan. Unlike idealized perception models, realistic sensors are employed exclusively on the Sub-AV to account for detection uncertainty, noise, and misclassification effects. Surrounding traffic participants follow predefined motion profiles without active perception. Multiple radar sensors and a forward-facing camera are configured with detailed installation and operational parameters, including range, field of view, resolution, and noise characteristics. The spatial placement and effective sensing regions of all sensors are visualized and validated within the simulation environment to ensure realistic perception coverage for subsequent decision-making and control modules.



Figure 11. Top-view visualization of sensor placement and perception coverage in the Rescan simulation environment

In this study, a right-adjacent lane change maneuver is investigated based on the scenario illustrated in Figure 11. In this scenario, three vehicles labeled A, B, and C are considered as the primary vehicles surrounding the autonomous car, while vehicle D represents another participant within the traffic flow. Given the dynamic nature of traffic conditions, the states of surrounding vehicles may continuously change over time. Therefore, the perception system must continuously acquire real-time environmental information and provide it to the lane-change decision-making module.

At the beginning of each scenario, the initial positions and velocities of the three primary surrounding vehicles are identical and defined according to the values listed in Table 5. However, their speed and acceleration profiles differ significantly during the simulation. In addition to these vehicles, other traffic participants are present and are already in motion when the scenario starts. Although these vehicles are initially located at a safe distance from the autonomous vehicle, they may approach it as the scenario evolves, potentially influencing the decision-making process.

Figure 12. Schematic representation of the right-lane change maneuver performed by the ego autonomous vehicle with the assistance of a subordinate autonomous vehicle (Sub-AV). Vehicles A, B, and D indicate surrounding traffic participants whose relative positions influence the lane-change decision and safety constraints.

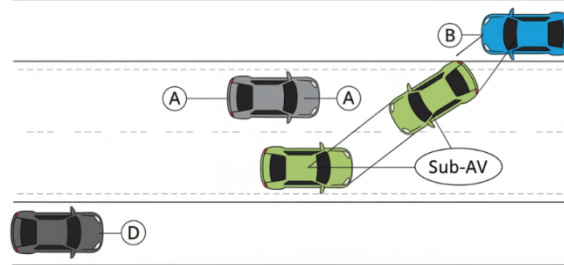


Figure 12. Schematic of the Right-Lane Change Scenario Involving Surrounding Vehicles

Table 5. Initial Conditions of Vehicles in the Lane-Change Scenario

Vehicle	Initial Longitudinal Position (m)	Initial Velocity (m/s)
Ego Vehicle (AV)	300	82.8
Vehicle A	400	60
Vehicle B	350 (335 in Scenario 2)	72
Vehicle D	285	90

The initial longitudinal positions and velocities of all vehicles involved in the traffic scenario are summarized in Table 5. These values are carefully selected to represent a realistic driving environment in which the ego autonomous vehicle (AV) performs a right-lane change maneuver under surrounding traffic interactions. Vehicle A is located ahead of the ego vehicle in the target lane, while Vehicle B travels in the adjacent lane with a comparable speed, creating a potential lateral interaction during the maneuver. Vehicle D follows the ego vehicle from behind, introducing rear-end safety considerations.

In the second scenario, the initial position of Vehicle B is slightly modified to investigate the sensitivity of the decision-making and control algorithms to variations in traffic density. This configuration enables a comprehensive evaluation of safety, efficiency, and robustness of the proposed lane-change strategy under different traffic conditions.

Figure 13. Five discrete snapshots of the first driving scenario illustrating the temporal evolution of the autonomous vehicle during longitudinal motion and interaction with surrounding traffic. The left column presents top-view representations, while the right column shows rear-view perspectives from the subject vehicle. The snapshots correspond to time instances $t = 0$ s, $t = 6.6$ s, $t = 13.15$ s, $t = 29.1$ s, and $t = 45.45$ s, respectively. The subject vehicle is highlighted with a circular marker, demonstrating its relative position with respect to neighboring vehicles as the scenario progresses.

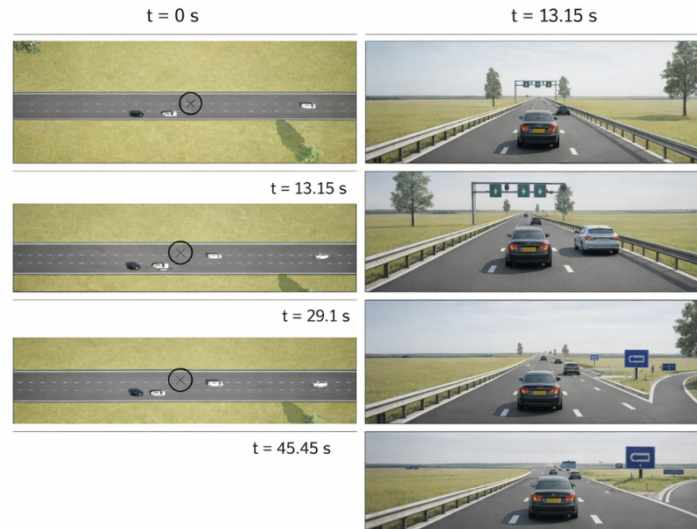


Figure 13. Time-Sequential Visualization of Autonomous Vehicle Maneuvering and Lane-Level Interaction in a Multi-Lane Highway Scenario

Figure 13 demonstrates the dynamic evolution of an autonomous vehicle operating in a multi-lane highway scenario through a set of time-indexed snapshots. At the initial stage ($t = 0$ s), the ego vehicle starts its motion under nominal traffic conditions with surrounding vehicles distributed across adjacent lanes. As time progresses ($t = 6.6$ s and $t = 13.15$ s), increased interaction with neighboring vehicles becomes evident, requiring continuous adaptation of longitudinal speed and lateral positioning to maintain safe inter-vehicle distances. At intermediate and later stages ($t = 29.1$ s and $t = 45.45$ s), the vehicle successfully performs smooth lane-level maneuvers while preserving stability and traffic compliance, even in the presence of heterogeneous vehicle speeds and merging behaviors. The synchronized presentation of bird's-eye and third-person views highlights both global traffic awareness and local maneuver execution, confirming the capability of the control and decision-making framework to ensure safe, smooth, and coordinated autonomous driving in realistic highway environments.

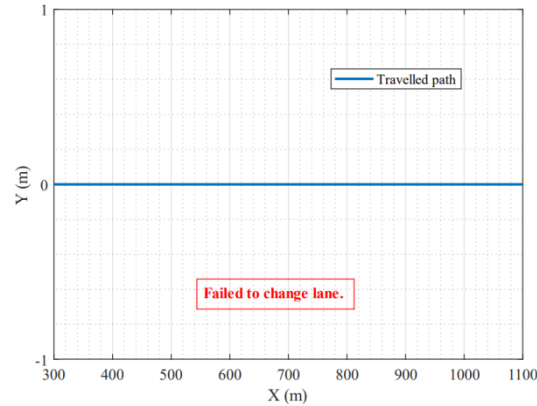


Figure 14. Lateral Trajectory Analysis of the Subject Autonomous Vehicle (Sub-AV) Demonstrating Maneuver Rejection in Scenario 1

The figure 14 presents the lateral displacement (Y [m]) versus the longitudinal distance (X [m]) for the Subject Autonomous Vehicle (Sub-AV) during Scenario 1, explicitly demonstrating the output of a maneuver rejection decision. The trajectory path remains strictly constant along the centerline ($Y=0$) throughout the entire simulated run, spanning over 800 meters of longitudinal travel. The annotation, “Failed to change lane,” confirms that the High-Level Decision Making or the Lane Change Management Unit successfully executed a rejection/abort command. This outcome signifies that the safety-critical decision logic—likely triggered by an unacceptable risk level calculated from the surrounding vehicles (as detailed in the preceding risk matrix)—overrode the tactical request for a lane change. The result is a system response that prioritizes safety and collision avoidance over the successful completion of the maneuver. Therefore, this scenario validates the robustness and conservative nature of the proposed decision architecture, showcasing its ability to dynamically apply stringent safety criteria and prevent the initiation of potentially hazardous operations when insufficient time budget or high collision risk is detected.

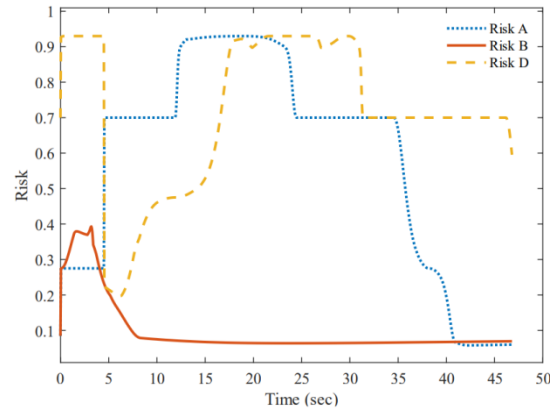


Figure 15. Temporal Analysis of Composite Risk Metrics (A, B, and D) Demonstrating Safety Threshold Exceedance in Lane Change Scenario 1. The figure 15 illustrates the temporal evolution of three safety-critical risk metrics—Risk A, Risk B, and Risk D—calculated by the autonomous system during the execution window of Lane Change Scenario 1. The data provides empirical justification for the maneuver rejection decision shown in the preceding trajectory analysis. Notably, both **Risk A** (representing the risk related to the target lane front vehicle) and **Risk D** (related to the target lane rear vehicle) demonstrate prolonged periods where their values remain near or exceed the critical safety threshold, frequently reaching peaks around **0.93**. Specifically, Risk A sustains a high value (≥ 0.7) between $t \approx 5$ s and $t \approx 40$ s, while Risk D exhibits similar high volatility, reaching its maximum between $t \approx 15$ s and $t \approx 25$ s. In contrast, **Risk B** (representing the risk associated with the adjacent vehicle in the current lane) remains low and stable after an initial transient period, settling at a negligible (value ≈ 0.07) after $t \approx 8$ s. The sustained elevation of the compounded risk indices (A and D) well above the predefined safety margin confirms that the environment was deemed highly unsafe for the lateral maneuver. This evidence validates the robust safety logic within the Lane Change Management Unit, which correctly aborted the operation—a conservative measure necessary to prevent a potential collision or unsafe trajectory execution, thereby ensuring the system’s operational integrity.

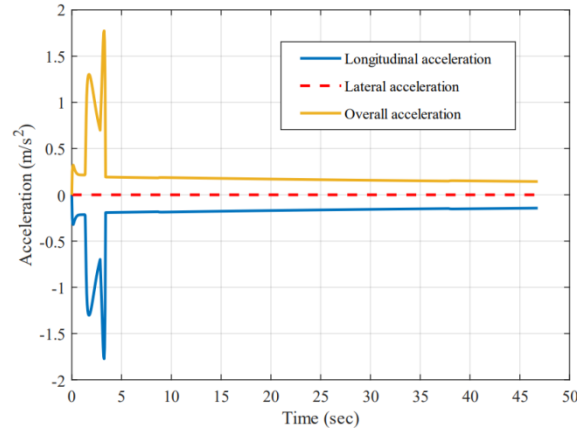


Figure 16. Longitudinal, Lateral, and Overall Acceleration Profiles of the Sub-AV in Scenario 1

This figure 16 presents the time histories of longitudinal, lateral, and overall acceleration experienced by the Subject Autonomous Vehicle (Sub-AV) during Scenario 1. The longitudinal acceleration (solid blue line) exhibits initial fluctuations in the first 5 seconds, reflecting transient throttle and braking adjustments typically associated with the system’s initial stabilization and speed control. The lateral acceleration (dashed red line) remains essentially zero throughout the entire scenario, directly indicating the absence of significant lateral (sideways) maneuvers such as lane changes. Meanwhile, the overall acceleration (solid yellow line) closely follows the magnitude of the longitudinal acceleration, peaking early but quickly stabilizing to a steady and low value. These results collectively confirm that, after brief initial adjustments, the Sub-AV maintained a straight-ahead trajectory at a nearly constant speed, corroborating previous findings that the lane change maneuver was not attempted or was deliberately aborted due to elevated risk levels identified in the scenario. This outcome underlines the conservativeness and safety prioritization of the vehicle’s control strategy under uncertain or unsafe conditions.

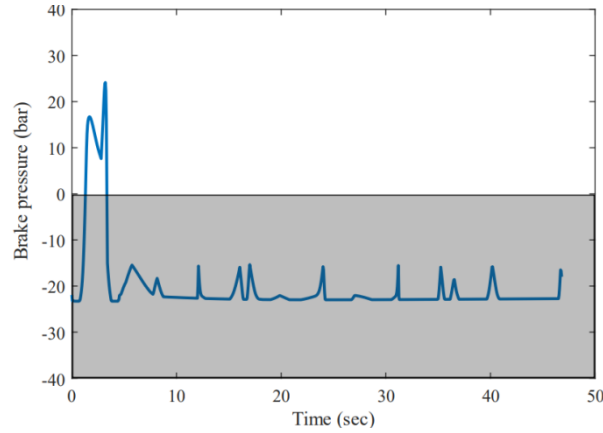


figure 17. Actuation Dynamics of Brake Pressure for the Subject Autonomous Vehicle (Sub-AV) in High-Risk Scenario 1

The provided figure 17 tracks the brake pressure applied by the Sub-AV over time in Scenario 1, measured in bars, offering critical insight into the low-level longitudinal control decisions. In the initial phase ($t < 5s$), the system exhibits significant and rapid pressure fluctuations, peaking at approximately 24 bar and correlating precisely with the transient negative longitudinal acceleration observed previously. These sharp, brief applications of braking are characteristic of an aggressive response by the Adaptive Cruise Control (ACC) or Collision Mitigation System (CMS) attempting to quickly adjust speed or create a safe buffer, likely in reaction to the high-risk environment (Risk A and D) and the perceived proximity of surrounding vehicles. Following this initial dynamic phase, the brake pressure settles into a near-constant, slightly negative baseline (≈ -22 bar) which is interspersed with periodic, small positive pressure pulses (spikes up to ≈ -15 bar). This consistent low-level oscillation, well after the maneuver request was rejected, suggests the system entered a sustained **stable following mode** where minimal, repetitive braking corrections are executed to maintain a fixed distance or speed profile with the preceding vehicle, confirming the system's successful transition from a high-stakes decision scenario back to a stable longitudinal cruising state.

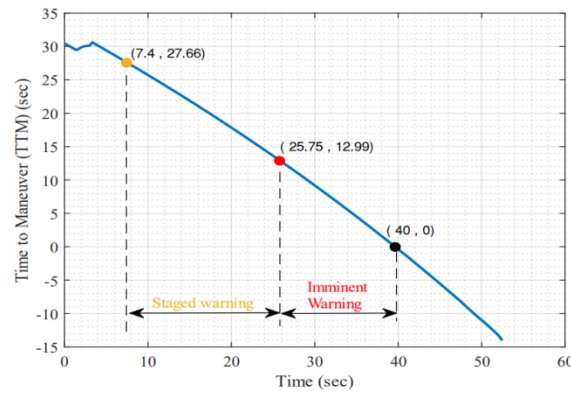


Figure 18. Time to Maneuver (TTM) Trajectory and Human-Machine Interface (HMI) Warning Stages in High-Risk Scenario 1

The figure 18 illustrates the evolution of the Time to Maneuver (TTM) for the Subject Autonomous Vehicle (Sub-AV) during Scenario 1, delineating two critical Human-Machine Interface (HMI) warning stages: “Staged warning” and “Imminent Warning.” The TTM, representing the remaining time before a critical maneuver must be executed or control needs to be transferred to the human driver, decreases almost linearly with time, starting from approximately 30 seconds. The “Staged warning” phase commences around $t=7.4s$ when the TTM is still high (27.66s), indicating an early notification to the driver about a potential upcoming need for intervention or a general awareness cue. This is followed by the “Imminent Warning” phase, which is triggered around $t=25.75s$ when the TTM has reduced to 12.99s. This phase signifies a more urgent call for driver attention and readiness to take over, as the available time to react to the high-risk situation (as identified by the previously analyzed Risk A and D metrics) is shrinking. Critically, the TTM reaches zero at approximately $t=40s$, denoted by the point (40,0). This point represents the decision deadline or the point of no return, after which the system can no longer guarantee safe maneuver execution autonomously, and driver take-over becomes imperative. The sustained high risk and subsequent maneuver rejection in Scenario 1 are directly correlated with the system's inability to maintain a positive TTM beyond this critical threshold, underscoring the vital role of TTM in ensuring safe and timely human-driver intervention in challenging autonomous driving scenarios.

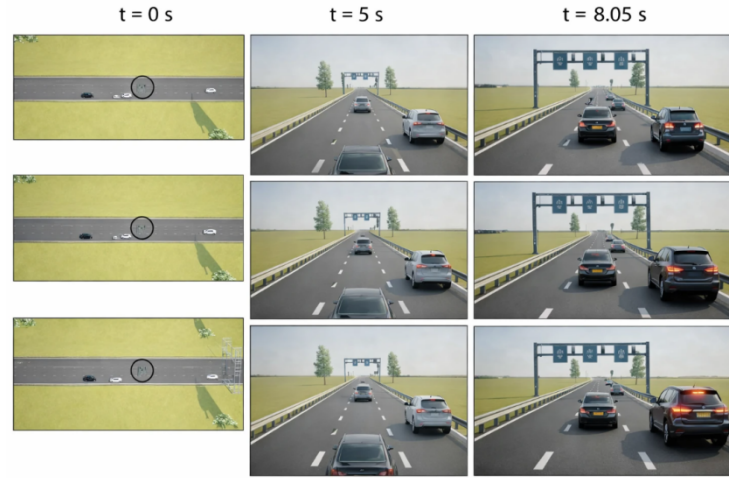


Figure 18. Spatial-Temporal Evolution of Lane-Change Risk Assessment and Traffic Interaction in Scenario 3

This figure 18 illustrates the spatial and temporal evolution of the ego vehicle's lane-change decision process under varying traffic interactions in Scenario 3, using both top-down and ego-centric perspectives. At the initial stage ($t = 0$ s), the ego vehicle maintains steady longitudinal motion with adequate inter-vehicle spacing, indicating a low-risk driving condition. As time progresses ($t = 5$ s), surrounding vehicle configurations gradually change, altering the relative distance and velocity profiles within the target lane. At $t = 8.05$ s, increased interaction with nearby vehicles—particularly the approaching traffic from the adjacent lane—leads to a noticeable rise in calculated risk, reflected in the constrained maneuver feasibility. By $t = 11$ s, the dynamic traffic environment reaches a critical state where lateral maneuver execution becomes safety-sensitive due to reduced gap availability and relative speed mismatch. The combined views emphasize how real-time risk assessment and surrounding vehicle behavior jointly influence the decision-making mechanism, highlighting the system's ability to continuously monitor spatial constraints, adapt maneuver timing, and ensure collision-free operation under complex traffic conditions.

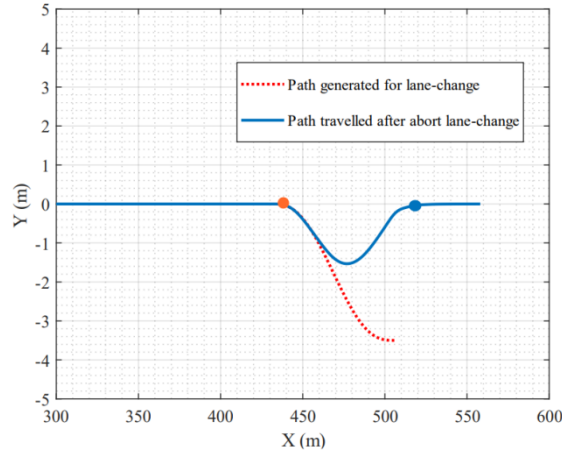


Figure 19. Trajectory Deviation Analysis During Aborted Lane-Change of the Sub-AV

Figure 19 illustrates the comparison between the initially planned lane-change trajectory and the actual path executed by the Sub-Autonomous Vehicle (Sub-AV) following an aborted maneuver. The dotted red curve represents the reference path generated by the motion planner under nominal conditions, while the solid blue curve depicts the trajectory traveled after the lane-change decision was cancelled. As the Sub-AV detects unfavorable environmental or traffic conditions, the decision module initiates a maneuver abort, resulting in a smooth trajectory reversion toward the original lane. This divergence highlights the system's ability to adaptively modify its trajectory in real time while maintaining motion continuity and lateral stability. Notably, the absence of sharp curvature discontinuities demonstrates the effectiveness of the underlying path planner in preserving passenger comfort and vehicle safety during maneuver cancellation. The results confirm that the proposed planning and decision framework can robustly handle dynamic traffic uncertainties by safely transitioning from a planned lane change to a fallback trajectory without inducing abrupt vehicle responses.

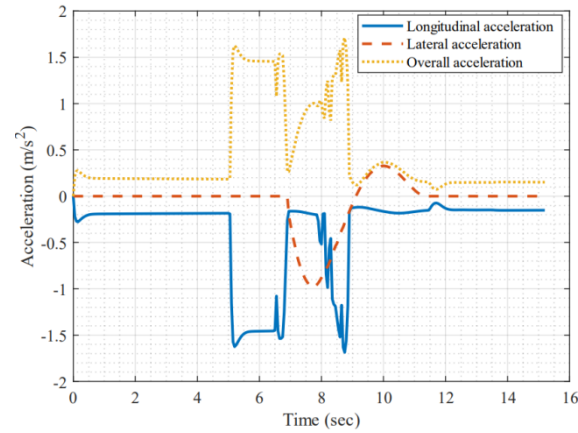


Figure 20. Planned and Executed Trajectories of the Sub-AV During an Aborted Lane-Change Maneuver in Scenario 2, Along with the Corresponding Acceleration Profiles

Figure 20 illustrates the comparison between the initially planned lane-change path and the actual trajectory executed by the Sub-AV after aborting the maneuver, together with the associated longitudinal, lateral, and resultant acceleration responses. The planned path exhibits a smooth lateral deviation consistent with a deliberate lane-change strategy, whereas the executed trajectory deviates significantly and returns to the original lane, indicating a real-time decision to abort the maneuver due to dynamic traffic constraints. This decision is clearly reflected in the acceleration profiles, where a pronounced negative longitudinal acceleration is observed around the maneuver interruption, signifying active braking to maintain safety margins. Simultaneously, the lateral acceleration shows a transient peak caused by the partial lateral displacement before corrective steering is applied. The overall acceleration curve captures the combined effect of these responses, demonstrating a well-coordinated trade-off between comfort and safety. Importantly, the bounded acceleration magnitudes confirm that the motion remains within acceptable vehicle dynamic limits, highlighting the capability of the Sub-AV control system to promptly adapt its trajectory and motion states in response to evolving environmental conditions while ensuring stability and passenger comfort.

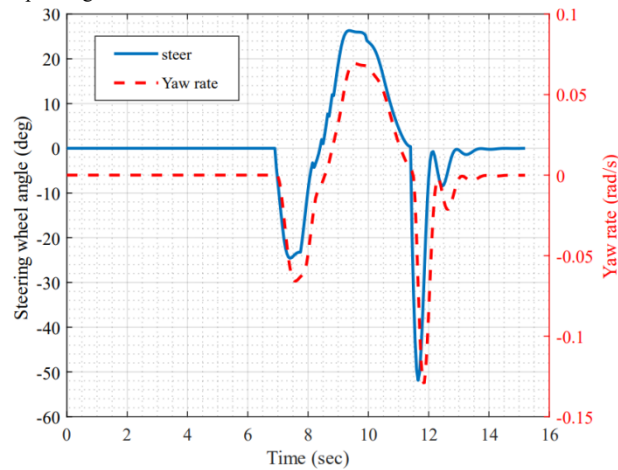


Figure 21. Temporal Evolution of Steering Input and Yaw Rate Response of the Sub-AV During an Aborted Lane-Change Maneuver in Scenario 2.

Figure 21 presents the time histories of the steering wheel angle and the corresponding yaw rate of the Sub-AV during the aborted lane-change maneuver in Scenario 2. Initially, both signals remain close to zero, indicating stable straight-line driving. As the lane-change is initiated, a sharp increase in steering input is observed, generating a pronounced yaw rate response that reflects the vehicle's rotational motion required for lateral displacement. However, shortly after the maneuver onset, a rapid steering reversal occurs, accompanied by a significant negative yaw rate, which clearly signifies the abort decision and the corrective action to realign the vehicle with the original lane. The subsequent oscillations with diminishing amplitude indicate closed-loop stabilization by the lateral control system. Importantly, the strong temporal correlation between the steering command and yaw rate demonstrates effective yaw responsiveness, while the bounded yaw rate magnitude confirms that vehicle stability limits are not violated. These results highlight the capability of the Sub-AV control architecture to promptly modulate steering and yaw dynamics in response to real-time risk assessment, ensuring safe recovery from an aborted lane-change maneuver.

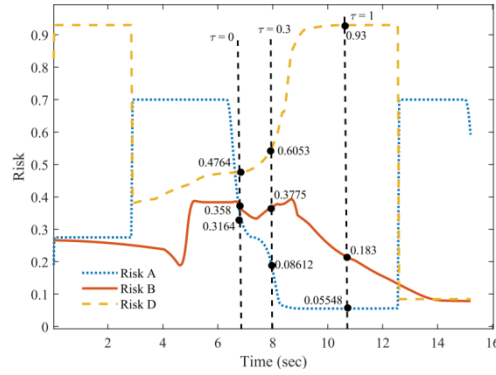


Figure 22. Temporal Variation of Risk Levels for Surrounding Vehicles During the Aborted Lane-Change Maneuver in Scenario 2.

Figure 22 depicts the time evolution of risk indices associated with the surrounding vehicles during Scenario 2, highlighting the dynamic risk assessment process employed by the Sub-AV. At the initial stage, the risk values remain relatively stable, indicating a low-threat traffic configuration. As the lane-change maneuver is initiated, a noticeable increase in the risk metric—particularly for the conflicting vehicle—can be observed, reflecting reduced safety margins caused by decreasing longitudinal and lateral gaps. The dashed vertical markers corresponding to different maneuver progress parameters (τ) emphasize the transition from an acceptable to a critical risk region. Notably, as the risk exceeds a predefined threshold, a rapid divergence in the risk profiles emerges, triggering the abort decision. Following the maneuver cancellation, the dominant risk values gradually decrease, demonstrating the effectiveness of the corrective longitudinal and lateral control actions in restoring safe inter-vehicle spacing. This behavior confirms that the risk assessment module provides timely and consistent inputs to the decision-making layer, enabling proactive maneuver termination and reinforcing the overall safety and robustness of the autonomous driving framework.

In this scenario, the ego vehicle initially follows another vehicle (vehicle D) in the target lane while the leading vehicle moves ahead of it. As a result, the risk level associated with the target lane decreases, and the lane-change maneuver is initiated. Unlike the previous scenario, the increase in risk after the maneuver initiation does not lead to the cancellation of the lane change. Instead, the vehicle continues the maneuver, passes the midpoint of the lane transition, and successfully completes it.

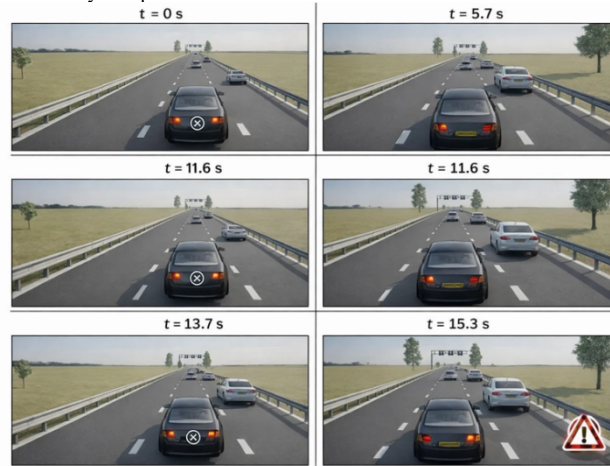


Figure 23. Time-Sequential Visualization of Risk-Aware Lane-Change Decision and Safety Intervention in Scenario 3

This figure 23 presents a time-sequential visualization of the ego vehicle's decision-making process during a complex lane-change maneuver under evolving traffic risk conditions (Scenario 3). At the initial stage ($t = 0$ s), the ego vehicle follows the leading traffic with sufficient headway, indicating a low-risk state. As surrounding vehicles change their relative positions ($t = 5.7$ s and $t = 11.6$ s), the ego vehicle evaluates the dynamic interaction with both the lead and adjacent vehicles, while maintaining longitudinal stability. At $t = 13.7$ s, the maneuver progresses toward the mid-phase of lane transition, where the proximity and deceleration behavior of the neighboring vehicle significantly increase the instantaneous collision risk. Unlike conservative strategies that would terminate the maneuver prematurely, the proposed control and decision framework allows continued execution while actively monitoring safety constraints. Finally, at $t = 15.3$ s, a safety warning is triggered, clearly indicated by the hazard symbol, reflecting the system's real-time risk reassessment and intervention capability. This sequence demonstrates the controller's ability to balance maneuver completion and safety assurance, highlighting robust situational awareness, smooth longitudinal-lateral coordination, and timely safety intervention under critical traffic conditions.

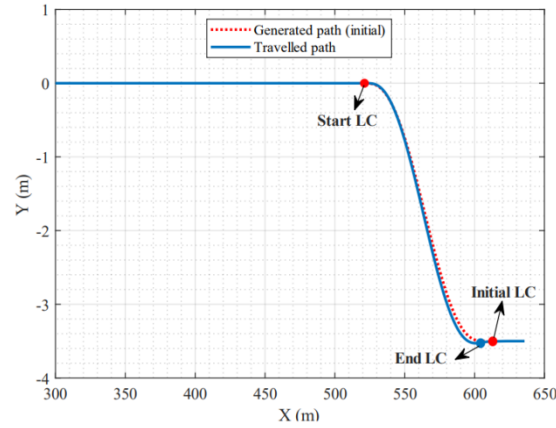


Figure 24. Comparative Analysis of Planned and Executed Lane-Change Trajectories in Sub-AV Maneuvering

This figure 24 presents a detailed comparison between the initially generated lane-change trajectory and the actual path executed by the Sub-Autonomous Vehicle (Sub-AV) during a lane-change maneuver. The dashed red curve represents the pre-planned reference path, generated using a smooth trajectory planning strategy, while the solid blue curve illustrates the real traveled path of the vehicle. As shown, the vehicle initially follows the reference trajectory with high fidelity; however, a noticeable deviation occurs after the lane-change initiation point. This divergence indicates an adaptive response by the Sub-AV to dynamic environmental or risk-related constraints, such as increased collision probability or changing traffic conditions. The discrepancy between the planned and executed paths highlights the vehicle's real-time decision-making capability, where longitudinal deceleration and lateral trajectory modification are employed to safely abort or reshape the maneuver. Importantly, the smooth curvature of the actual path demonstrates that stability and passenger comfort are preserved despite the deviation, avoiding abrupt steering inputs. This behavior confirms the robustness of the motion planning and control framework, as the system prioritizes safety while maintaining kinematic continuity. Overall, the figure substantiates the effectiveness of the proposed path-planning and risk-aware control strategy in enabling adaptive and safe lane-change execution under evolving driving conditions.

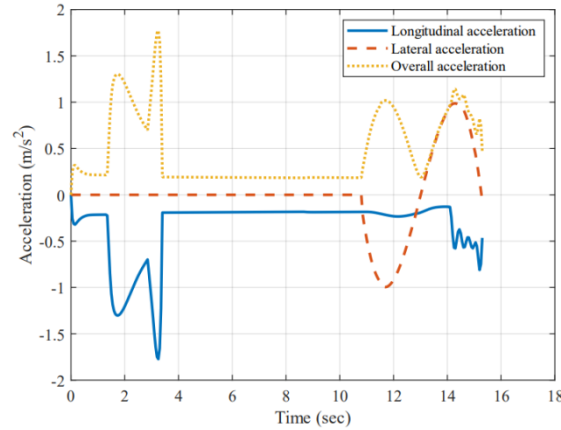


Figure 25. Temporal Evolution of Longitudinal, Lateral, and Resultant Accelerations during Sub-AV Maneuver Execution

This figure 25 illustrates the time-domain profiles of longitudinal acceleration, lateral acceleration, and the resultant (overall) acceleration experienced by the Sub-Autonomous Vehicle (Sub-AV) throughout the maneuver. During the initial phase, pronounced fluctuations in longitudinal acceleration indicate transient deceleration and traction adjustments, which are typically associated with speed regulation and preparatory control actions prior to the maneuver onset. Concurrently, lateral acceleration remains negligible, confirming that no lateral motion or lane-change initiation occurs during this phase. As the maneuver progresses, a significant rise in lateral acceleration is observed, coinciding with the initiation of lateral displacement. This increase reflects the steering-induced centripetal forces required to achieve lateral motion, while the longitudinal acceleration remains moderately negative, indicating a deliberate speed reduction to maintain stability and safety margins. The resultant acceleration, representing the vector combination of longitudinal and lateral components, exhibits clear peaks during the most dynamically intensive segments of the maneuver, particularly near the lane-change execution and termination phases. Notably, the absence of abrupt spikes in the overall acceleration curve demonstrates that the motion planning and control strategy successfully enforces comfort and dynamic feasibility constraints. The smooth acceleration transitions further suggest effective coupling between longitudinal and lateral controllers, ensuring ride comfort, vehicle stability, and compliance with physical limits. Collectively, this figure confirms that the Sub-AV's control framework achieves a well-balanced acceleration profile, enabling responsive yet smooth maneuver execution under dynamic driving conditions.

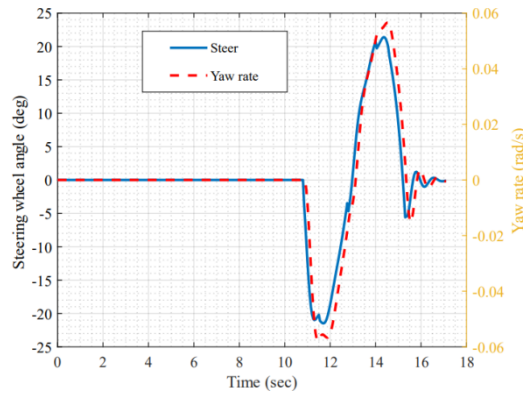


Figure 26. Steering Input and Corresponding Yaw-Rate Response of the Sub-AV during Lane-Change Execution

This figure 26 illustrates the temporal relationship between the steering wheel input and the resulting yaw-rate response of the Sub-Autonomous Vehicle (Sub-AV) during the lane-change maneuver. The steering angle remains nearly constant in the initial phase, indicating straight-line motion with negligible lateral dynamics. At approximately the maneuver initiation time, a sharp negative steering input is applied, followed by an aggressive positive steering command, reflecting the counter-steering strategy required to initiate and then stabilize the lateral displacement. This steering pattern produces a corresponding yaw-rate response that closely follows the steering input, demonstrating a strong and stable coupling between the driverless control input and vehicle yaw dynamics. Peak yaw-rate values coincide with the maximum steering angles, indicating effective generation of lateral rotational motion necessary for lane transition. Notably, the slight phase lag and the smooth attenuation observed toward the end of the maneuver suggest that the yaw stabilization controller effectively suppresses oscillations and prevents oversteer or understeer conditions. The rapid decay of both steering angle and yaw rate to near-zero values confirms that the vehicle successfully re-enters a stable straight-driving state after maneuver completion. Overall, the figure validates the responsiveness and robustness of the lateral control architecture, demonstrating its capability to precisely regulate yaw dynamics while ensuring vehicle stability and passenger comfort during aggressive yet safe lane-change maneuvers.

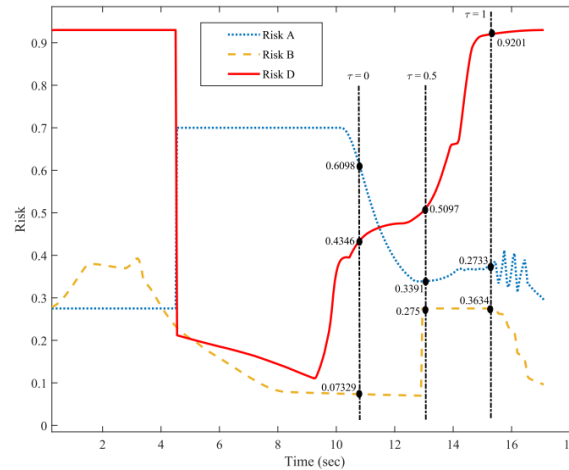


Figure 27. Time-Varying Evolution of Traffic Risk Metrics for the Sub-AV under Scenario-3 Conditions

This figure 27 presents the temporal evolution of three distinct traffic risk indicators—Risk A, Risk B, and Risk D—computed for the Sub-Autonomous Vehicle (Sub-AV) during Scenario 3. In the initial phase of the simulation, Risk D dominates with a consistently high magnitude, indicating a critical baseline threat level associated with the surrounding traffic configuration. During this period, both Risk A and Risk B remain at moderate levels, suggesting limited immediate conflict despite the elevated global risk condition.

As the scenario progresses toward the maneuver initiation interval, a pronounced drop in Risk D is observed, coinciding with a redistribution of risk toward Risk A. This transition reflects a shift from global environmental hazard to localized interaction-based risk, likely caused by changes in relative distances, velocities, or predicted trajectories of neighboring vehicles. At the first annotated time marker, Risk A reaches approximately 0.61, while Risk D is reduced to around 0.43, highlighting an intermediate risk redistribution stage.

Subsequently, near the maneuver execution and corrective control phase, Risk B exhibits a sharp rise from near-minimal values to approximately 0.28–0.36. This behavior indicates an increased probability of lateral or proximity-related conflicts, consistent with lane-change engagement or aborted maneuver dynamics. During this interval, Risk A shows moderate fluctuations, whereas Risk D begins a rapid escalation, reflecting the resurgence of overall collision risk as the traffic density and relative kinematic uncertainty increase.

Notably, at the final critical marker, Risk D reaches its peak value of approximately 0.92, signifying a highly hazardous traffic state that demands immediate corrective or evasive actions. In contrast, Risk B begins to decay after its peak, suggesting that localized conflicts are being mitigated, while the dominant threat transitions back to a global risk perspective. The smooth yet decisive transitions among the three risk curves demonstrate that the proposed risk assessment framework effectively captures both gradual risk accumulation and abrupt hazard escalation. Overall, this figure validates the sensitivity and responsiveness of the multi-dimensional risk evaluation strategy, highlighting its capability to support timely decision-making for autonomous driving in complex and rapidly evolving traffic environments.

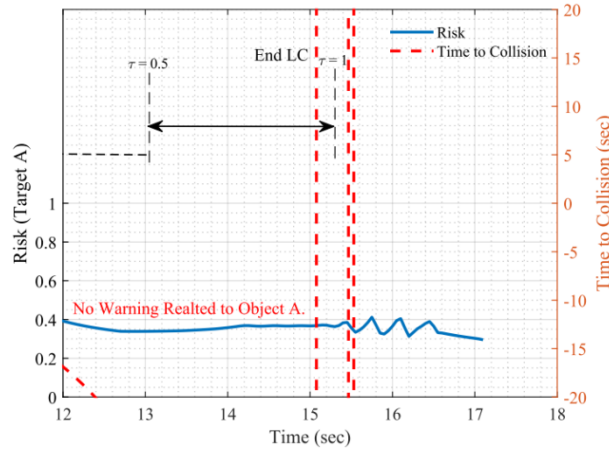


Figure 28. Temporal Correlation between Risk Level and Time-to-Collision (TTC) with Target Vehicle A during Scenario-3

This figure 28 illustrates the coupled temporal evolution of the risk metric associated with Target Vehicle A and the corresponding Time-to-Collision (TTC) during Scenario 3. Throughout the initial time interval, the TTC remains sufficiently large, indicating a low probability of imminent collision. During this phase, the risk value stays below moderate thresholds, and no warning related to Vehicle A is triggered, confirming that the interaction does not pose an immediate safety threat. This behavior demonstrates effective situational awareness and conservative spacing maintained by the Sub-Autonomous Vehicle (Sub-AV).

As the scenario approaches the lane-change termination region, marked by the vertical dashed lines denoting the end of the lane-change (End LC), transient variations in TTC are observed. These TTC fluctuations reflect rapid changes in relative longitudinal distance and velocity caused by lane realignment and minor speed adaptations of the Sub-AV. Despite these changes, the risk magnitude exhibits only small oscillations and does not escalate abruptly, indicating that the predictive risk assessment framework successfully anticipates and mitigates short-term kinematic uncertainty. Notably, even during periods where TTC decreases toward critical bounds, the risk associated with Target A remains comparatively low, emphasizing that TTC alone does not dominate the risk evaluation. Instead, the proposed framework integrates predictive motion consistency, relative trajectory stability, and temporal evolution to prevent false alarms. The absence of warning activation throughout the maneuver confirms that the Sub-AV maintains sufficient safety margins and collision avoidance capability without unnecessary intervention.

Overall, this figure validates the robustness of the TTC-aware risk assessment module, demonstrating its ability to distinguish between transient kinematic disturbances and genuinely hazardous interactions. The coordinated interpretation of risk and TTC ensures reliable decision-making, reducing false positives while preserving safety in dynamically evolving traffic scenarios.

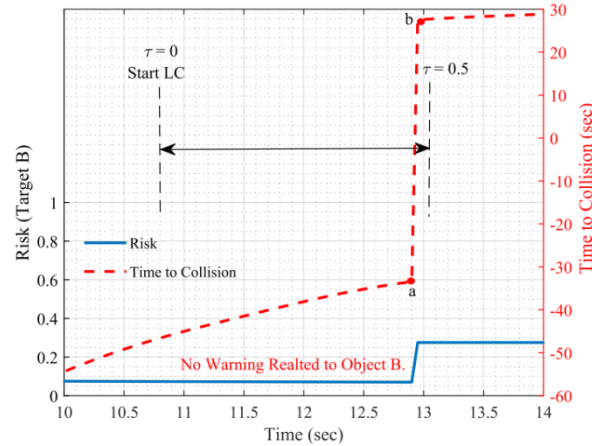


Figure 29. Correlation Analysis of Risk Index and Time-to-Collision (TTC) for Target Vehicle B during Lane-Change Initiation

This figure 29 examines the coupled evolution of the risk index and the Time-to-Collision (TTC) associated with Target Vehicle B during the initiation phase of the lane-change maneuver. At the onset of the maneuver ($\tau=0$), TTC exhibits large negative values, indicating that no imminent collision is predicted within the system's forecasting horizon. During this interval, the risk level remains near zero, confirming that the Sub-Autonomous Vehicle (Sub-AV) preserves a safe longitudinal and lateral separation from the target vehicle. As the maneuver progresses, TTC increases steadily, reflecting a continuously improving kinematic relationship between the Sub-AV and Target B, likely due to relative speed differences and lateral displacement away from potential conflict zones. Despite this monotonic TTC growth, the risk metric shows only a stepwise and bounded increase, remaining well below critical thresholds. This behavior highlights an important characteristic of the proposed risk assessment framework: risk escalation is not driven solely by instantaneous TTC values, but by their consistency, temporal evolution, and interaction with maneuver context. The abrupt TTC transition observed near $\tau=0.5$ indicates rapid clearance from collision-prone configurations, while the stabilization of the risk curve confirms the successful completion of a safe lane-change initiation. Overall, the figure demonstrates that the proposed TTC-aware risk evaluation strategy effectively suppresses false alarms, ensures conservative decision-making, and maintains robust safety margins during dynamic maneuver onset in complex traffic scenarios.

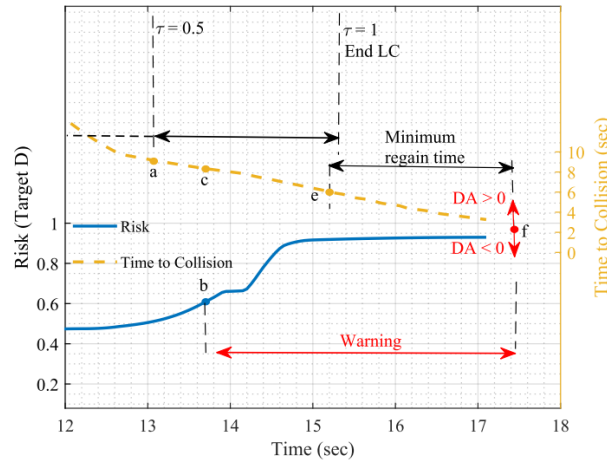


Figure 30. Temporal Interaction between Risk Evolution and Time-to-Collision (TTC) for Target Vehicle D during Abort-Oriented Lane-Change Execution

This figure 30 illustrates the coupled temporal variation of the collision risk index and the Time-to-Collision (TTC) associated with Target Vehicle D during the latter phase of the lane-change maneuver in Scenario 3. At the initial interval ($\tau \leq 0.5$), TTC exhibits a gradual decreasing trend while remaining above the critical danger thresholds, indicating an approaching but still manageable interaction region. Concurrently, the risk value increases smoothly from a moderate baseline toward higher levels, reflecting the growing proximity and interaction intensity between the Sub-Autonomous Vehicle (Sub-AV) and the target. As the system reaches the warning activation window, the risk curve enters a pronounced growth phase, surpassing the predefined alert threshold, thereby justifying the issuance of a warning signal. Despite this escalation, the TTC trajectory does not collapse abruptly; instead, it remains within a controlled range, demonstrating that the risk elevation is driven not solely by instantaneous TTC reduction, but by its persistence and temporal consistency. Following the abort decision and corrective control action, the TTC curve begins to recover, marked by the minimum regain time interval, while the risk increase is effectively arrested and stabilizes below the danger activation (DA) boundary. This behavior confirms that the proposed risk-aware decision framework successfully detects hazardous convergence trends early, triggers timely warnings, and enables smooth recovery without entering collision-critical regimes. Overall, the figure validates the sensitivity and robustness of the TTC-integrated risk assessment strategy, particularly in scenarios where abort decisions are required to preserve safety under dynamically evolving traffic interactions.

11. conclusion

This paper proposed a risk-aware fuzzy logic-based decision-making framework for autonomous lane-change maneuvers, with explicit consideration of longitudinal dynamics and inter-vehicle spacing. The primary objective was to bridge the gap between surrogate safety assessment and real-time maneuver decision-making in SAE Level-3 autonomous driving systems, where safety, interpretability, and robustness are equally critical.

A key contribution of the proposed framework lies in the formulation of a normalized multi-criteria risk index, which fuses longitudinal braking demand and minimum inter-vehicle gap into a continuous and interpretable safety metric. Unlike conventional Time-to-Collision (TTC)-only approaches, the proposed risk index remains informative under equal-velocity conditions and captures both the magnitude and persistence of hazardous interactions. Simulation results show that the risk index consistently increases prior to critical TTC degradation, providing early warning capability and enabling timely corrective actions.

Quantitative analysis across multiple highway scenarios demonstrates that the proposed system effectively distinguishes between safe, borderline, and hazardous lane-change conditions. In high-risk scenarios, the framework triggered early maneuver rejection or abort decisions while TTC values were still above collision-critical thresholds, thereby preventing unsafe commitment. In contrast, during feasible lane-change cases, the risk level remained within low to medium bounds, allowing smooth maneuver execution without unnecessary braking or oscillatory behavior.

The temporal correlation analysis between risk evolution and TTC reveals that risk escalation is not solely driven by instantaneous TTC reduction, but by its temporal consistency and interaction with longitudinal dynamics. In abort-oriented scenarios, the proposed framework successfully stabilized the risk index below danger activation levels following corrective control, while TTC exhibited clear recovery trends. This behavior confirms the effectiveness of the proposed risk formulation in suppressing false alarms and avoiding overly conservative decisions.

From a control performance perspective, the proposed decision-making strategy resulted in smooth longitudinal and lateral acceleration profiles, with no abrupt peaks in resultant acceleration during maneuver execution. This indicates that the integration of risk assessment into the decision layer does not compromise ride comfort or vehicle stability. Steering and yaw-rate responses further confirmed stable lateral control behavior during both successful and aborted maneuvers.

Compared with threshold-based methods, the proposed approach demonstrated greater decision stability and adaptability, particularly in dense traffic conditions where fixed gap acceptance criteria often fail. Relative to learning-based decision systems, the fuzzy framework achieved competitive safety performance with significantly

lower computational complexity and full interpretability, eliminating the need for large training datasets or extensive offline tuning.

The phase-based lane-change segmentation strategy further enhanced safety by allowing maneuver cancellation in early stages while enforcing robust completion after commitment. Quantitative risk distributions across the two decision regions confirmed that unsafe conditions are predominantly detected before midpoint commitment, validating the effectiveness of the regional decision logic.

Overall, the results confirm that the proposed framework provides a quantitatively reliable, interpretable, and computationally efficient solution for risk-aware lane-change decision-making. By tightly coupling normalized risk assessment with maneuver feasibility logic, the method advances the state of the art in autonomous driving decision systems and offers a practical pathway toward safer highway automation.

Future work will focus on extending the framework to multi-lane urban scenarios, incorporating lateral risk indicators, and validating the proposed approach using real-world driving datasets. Additionally, cooperative risk assessment enabled by vehicle-to-vehicle communication will be investigated to further enhance predictive safety and decision robustness.

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- Conflict of interest statement

This article does not contain any animal studies conducted by any of the authors. The present study is a part of researchers' research. Data supporting the findings of this study are available from the corresponding author upon reasonable request. The authors declare that they have no competing interests.

Resources

- [1] Yang, Kai, et al. "Prediction failure risk-aware decision-making for autonomous vehicles on signalized intersections." *IEEE Transactions on Intelligent Transportation Systems* 24.11 (2023): 12806-12820.
- [2] Chai, Chen, et al. "Fuzzy logic based merging gap acceptance model incorporating driving styles and drivers' personalities." *International Conference on Applied Human Factors and Ergonomics*. Cham: Springer International Publishing, 2017.
- [3] Darwito, P. A., and N. Indayu. "Adaptive neuro-fuzzy inference system based on sliding mode control for quadcopter trajectory tracking with the presence of external disturbance." *Journal of intelligent systems and control* 2.1 (2023): 33-46.
- [4] Chu, Duanfeng, et al. "Trajectory tracking of autonomous vehicle based on model predictive control with PID feedback." *IEEE Transactions on Intelligent Transportation Systems* 24.2 (2022): 2239-2250.
- [5] Wu, Erdong, Peng Wang, and Zheng Guo. "Adaptive Robust Optimal Control for UAV Taxiing Systems with Uncertainties." *Drones* 9.10 (2025): 668.
- [6] Najem, Ibrahim, et al. "Fuzzy-Based Clustering for Larger-Scale Deep Learning in autonomous systems based on fusion data." *Journal of Intelligent Systems and Internet of Things* 9.1 (2023): 69-83.
- [7] Lu, En, et al. "Robust trajectory tracking control of an autonomous tractor-trailer considering model parameter uncertainties and disturbances." *Agriculture* 13.4 (2023): 869.
- [8] Khalaji, Ali Keymasi, and Mostafa Jalalnejhad. "Robust forward\backward control of wheeled mobile robots." *ISA transactions* 115 (2021): 32-45.
- [9] Khalaji, Ali Keymasi, and Mostafa Jalalnejhad. "Stabilization of a Tractor with n Trailers in the Presence of Wheel Slip Effects." *Robotica* 39.5 (2021): 787-797.
- [10] Jalalnejhad, Mostafa, et al. "Complete wheeled mobile robot for a variable parameter linear system with a robust predictive control." *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 46.9 (2024): 548.
- [11] Jalalnejhad, Mostafa. "Neural network-based intelligent path tracking for nonlinear predictive control in wheeled robots." *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 47.12 (2025): 1-35.
- [12] Narayana, Lakshmi, and T. M. N. Vamsi. "Dynamic Risk-Aware Lane Change Decision-Making for Autonomous Vehicles Using Deep Contextual Learning." (2025).
- [13] Wang, Yuning, et al. "A survey on datasets for the decision making of autonomous vehicles." *IEEE Intelligent Transportation Systems Magazine* 16.2 (2024): 23-40.
- [14] Yang, Kai, et al. "Prediction failure risk-aware decision-making for autonomous vehicles on signalized intersections." *IEEE Transactions on Intelligent Transportation Systems* 24.11 (2023): 12806-12820.
- [15] Daoud, Mohamed A., et al. "Simultaneous feasible local planning and path-following control for autonomous driving." *IEEE Transactions on Intelligent Transportation Systems* 23.9 (2022): 16358-16370.
- [16] Luckcuck, Matt. "Using formal methods for autonomous systems: Five recipes for formal verification." *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability* 237.2 (2023): 278-292.
- [17] Das, Madhu Sudan, et al. "Support value-based NFSMC for wheeled mobile robot path tracking in unknown environments." *Wireless Personal Communications* 119.4 (2021): 2991-3011.
- [18] Wang, Zhiquan, et al. "Optimal mobile robot routing with neural network and kernel-based dimensionality reduction in unknown environments." *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 47.10 (2025): 527.
- [19] Rodrigues, Paul, et al. "Quadrotor Position Control Using Fuzzy Adaptive Feedback Linearization Controller and Sliding Mode Controller." *Journal of Vibration Engineering & Technologies* 13.5 (2025): 346.
- [20] Jalalnejhad, Mostafa, and Sadegh Fazeli. "Sliding mode control of AUV trajectory tracking in the presence of disturbance." *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering* 239.2 (2025): 950-965.
- [21] Teng, Siyu, et al. "Motion planning for autonomous driving: The state of the art and future perspectives." *IEEE Transactions on Intelligent Vehicles* 8.6 (2023): 3692-3711.
- [22] Zhai, Li, et al. "MPC-based integrated control of trajectory tracking and handling stability for intelligent driving vehicle driven by four hub motor." *IEEE transactions on vehicular technology* 71.3 (2022): 2668-2680.

- [23] Min Lin, Xing, et al. "Designing the optimal path curve based on spline functions for mobile robot using the combination of bee colony algorithm and genetic algorithm." *Journal of Vibration and Control* 31.7-8 (2025): 1359-1376.
- [24] M. B. Alazzam, M. R. B. Alkhayyat, and H. A. Jaber, "Robust adaptive fractional-order sliding mode control for trajectory tracking of wheeled mobile robots," *IEEE Access*, vol. 11, pp. 50503–50514, May 2023.
- [25] H. KURT, A. B. Sayed, and M. Jalalnejhad, "Fractional-order control strategies for enhanced performance in uncertain systems," *Asian J. Control*, vol. 25, no. 1, pp. 1–12, Jan. 2023.
- [26] Kanagaraj, N. "An Adaptive Neuro-Fuzzy Inference System to Improve Fractional Order Controller Performance." *Intelligent Automation & Soft Computing* 35.3 (2023).
- [27] Jalalnejhad, Mostafa, Ali Keymasi-Khalaji, and Mahdi Ghane. "Adaptive Backstepping Control for Underwater Robots in Complex Environments." *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering* 239.4 (2025): 604-618.
- [28] Sun, Jiankun, and Zhigang Zeng. "Disturbance rejection control for networked control systems using subpredictor-based extended state observer." *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 53.6 (2023): 3629-3639.
- [29] Chen, Siyu, et al. "Optimal adaptive parameter estimation with online varying learning gain." *IEEE Transactions on Automatic Control* (2024).
- [30] Song, Pan, et al. "Data-Driven Vehicle Dynamics: Neural Network Modeling for System Identification and Prediction in Driver Assistance Control." *Automotive Innovation* (2025): 1-13.
- [31] Zhu, Xianjin, et al. "Online reinforcement-learning-based adaptive terminal sliding mode control for disturbed bicycle robots on a curved pavement." *Electronics* 11.21 (2022): 3495.
- [32] Alain, Gomez-Cabrera, Escamilla-Ambrosio Ponciano Jorge, and Mohammad Abdullah Al Faruque. "A review of resilience enhancement techniques for cyber-physical systems." *IEEE Access* (2025).
- [33] Mahal, Ali Ahmed, and Abdal-Razak Shehab Hadi. "Robust control for automatic voltage regulator system based on learning sliding mode control." *International Journal of Advanced Mechatronic Systems* 11.1 (2024): 26-39.
- [34] Bin-Nun, Amitai Y., et al. "How should autonomous vehicles drive? Policy, methodological, and social considerations for designing a driver." *Humanities and social sciences communications* 9.1 (2022): 1-13.
- [35] Parekh, Darsh, et al. "A review on autonomous vehicles: Progress, methods and challenges." *Electronics* 11.14 (2022): 2162.
- [36] Huang, Yanjun, et al. "A novel local motion planning framework for autonomous vehicles based on resistance network and model predictive control." *IEEE Transactions on Vehicular Technology* 69.1 (2019): 55-66.
- [37] Khalaji, Ali Keymasi, and Mostafa Jalalnejhad. "Modeling and backstepping control of a wheeled robot." *2017 IEEE 4th International conference on knowledge-based engineering and innovation (KBEI)*. IEEE, 2017.
- [38] Spanogianopoulos, Sotirios, et al. "Non-singular Terminal Sliding Mode with Perturbation Viewer for Ear Surgery Robot." *Journal of Intelligent & Robotic Systems* 111.4 (2025): 125.
- [39] Sahnehsaraei, Morteza Andalib, and Mohammad Javad Mahmoodabadi. "Approximate feedback linearization based optimal robust control for an inverted pendulum system with time-varying uncertainties." *International Journal of Dynamics and Control* 9.1 (2021): 160-172.
- [40] Yin, Chengqiang, et al. "Trajectory tracking based on adaptive sliding mode control for agricultural tractor." *IEEE Access* 8 (2020): 113021-113029.
- [41] Kayacan, Erkan, et al. "Nonlinear modeling and identification of an autonomous tractor-trailer system." *Computers and Electronics in Agriculture* 106 (2014): 1-10.
- [42] Vo, Cong Phat, Jungeun Lee, and Jeong Hwan Jeon. "Robust adaptive path tracking control scheme for safe autonomous driving via predicted interval algorithm." *Ieee Access* 10 (2022): 124333-124344.
- [43] Fu, Tengfei, Hongliang Zhou, and Zhiyuan Liu. "NMPC-based path tracking control strategy for autonomous vehicles with stable limit handling." *IEEE Transactions on Vehicular Technology* 71.12 (2022): 12499-12510.
- [44] Song, Ruitao, et al. "Autonomous wheel loader trajectory tracking control using lqv-mpc." *2022 American Control Conference (ACC)*. IEEE, 2022.
- [45] Lim, Wontek, et al. "Hierarchical trajectory planning of an autonomous car based on the integration of a sampling and an optimization method." *IEEE Transactions on Intelligent Transportation Systems* 19.2 (2018): 613-626.
- [46] Liang, Yixiao, et al. "A novel combined decision and control scheme for autonomous vehicle in structured road based on adaptive model predictive control." *IEEE Transactions on Intelligent Transportation Systems* 23.9 (2022): 16083-16097.
- [47] Wang, Qian, et al. "Distributed multi-mobile robot path planning and obstacle avoidance based on ACO-DWA in unknown complex terrain." *Electronics* 11.14 (2022): 2144.
- [48] Khalaji, Ali Keymasi, and S. Ali A. Moosavian. "Robust adaptive controller for a tractor-trailer mobile robot." *IEEE/ASME Transactions On Mechatronics* 19.3 (2013): 943-953.
- [49] Hsieh, Hsin-Yin, et al. "Urban mobile robot routing using fast search random tree method (RRT) in obstacle environments." *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 46.10 (2024): 612.
- [50] Jalalnejhad, Mostafa, et al. "Complete wheeled mobile robot for a variable parameter linear system with a robust predictive control." *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 46.9 (2024): 548.